



Review

Cellulosic and hemicellulosic fractions of sugarcane bagasse: Potential, challenges and future perspective

Alokika^{a,1}, Anu^{a,1}, Anil Kumar^b, Vinod Kumar^c, Bijender Singh^{a,d,*}^a Laboratory of Bioprocess Technology, Department of Microbiology, Maharshi Dayanand University, Rohtak 124001, Haryana, India^b Department of Botany, Pt. N.R.S. Govt. College, Rohtak 124001, Haryana, India^c Department of Chemistry, Central University of Haryana, Jant-Pali, Mahendergarh 123031, Haryana, India^d Department of Biotechnology, Central University of Haryana, Jant-Pali, Mahendergarh 123031, Haryana, India

ARTICLE INFO

Article history:

Received 1 October 2020

Received in revised form 29 November 2020

Accepted 23 December 2020

Available online 29 December 2020

Keywords:

Sugarcane bagasse

Cellulose

Hemicellulose

Delignification

Pretreatment

Value-added products

ABSTRACT

Sugarcane bagasse is a rich source of cellulose (32–45%), hemicellulose (20–32%) and lignin (17–32%), 1.0–9.0% ash and some extractives. Huge amount of the generation of sugarcane bagasse has been a great challenge to industries and environment at global level for many years. Though cellulosic and hemicellulosic fractions in bagasse makes it a potential raw substrate for the production of value-added products at large scale, the presence of lignin hampers its saccharification which further leads to low yields of the value-added products. Therefore, an appropriate pretreatment strategy is of utmost importance that effectively solubilizes the lignin that exposes cellulose and hemicellulose for enzymatic action. Pretreatment also reduces the biomass recalcitrance i.e., cellulose crystallinity, structural complexity of cell wall and lignification for its effective utilization in biorefinery. Sugarcane bagasse served as nutrient medium for the cultivation of diverse microorganisms for the production of industrially important metabolites including enzymes, reducing sugars, prebiotic, organic acids and biofuels. Sugarcane bagasse has been utilized in the generation of electricity, syngas and as biosorbent in the bioremediation of heavy metals. Furthermore, the ash generated from bagasse is an excellent source for the synthesis of high strength and light weight bricks and tiles. Present review describes the utility of sugarcane bagasse as sustainable and renewable lignocellulosic substrate for the production of industrially important multifarious value-added products.

© 2020 Elsevier B.V. All rights reserved.

Contents

1.	Introduction	565
2.	World scenario for occurrence of sugarcane bagasse	567
3.	Composition, mechanical properties and structure of sugarcane bagasse.	567
3.1.	Composition	567
3.2.	Mechanical properties	567
3.3.	Structure	567
4.	Pretreatment strategies for sugarcane bagasse	567
4.1.	Physical pretreatments.	568
4.1.1.	Mechanical pretreatment	568
4.1.2.	Ultrasound pretreatment	568
4.1.3.	Microwave pretreatment	568
4.2.	Chemical pretreatments	569
4.2.1.	Acid pretreatment.	569
4.2.2.	Alkaline pretreatment	569
4.2.3.	Ionic liquids pretreatment	569
4.3.	Physiochemical pretreatment.	569

* Corresponding author at: Laboratory of Bioprocess Technology, Department of Microbiology, Maharshi Dayanand University, Rohtak 124001, Haryana, India.

E-mail addresses: ohlanbs@gmail.com, ohlanbs@cuh.ac.in (B. Singh).¹ Both contributed equally

4.3.1.	Steam explosion	569
4.3.2.	Organosolv pretreatment	569
4.3.3.	Hot water pretreatment	570
4.3.4.	Ammonia fibre explosion	570
4.4.	Biological pretreatment	570
5.	Value-added products from sugarcane bagasse	570
5.1.	Biofuel production	570
5.1.1.	Biohydrogen	570
5.1.2.	Bioethanol	570
5.1.3.	Biobutanol	571
5.1.4.	Xylitol	572
5.2.	Electricity generation	573
5.3.	As animal feed	573
5.4.	Industrial enzymes	574
5.4.1.	Xylanase	574
5.4.2.	Cellulase	574
5.4.3.	α -Amylase	574
5.4.4.	Lipase	574
5.5.	Sugars and oligosaccharides production	574
5.6.	Used in gasification in gasifier system	575
5.7.	As biosorbent for removal of hazardous material	575
5.8.	Sugarcane bagasse ash for stabilized earth blocks and tiles.	576
5.9.	Commercial products from sugarcane bagasse	576
5.9.1.	Single cell protein	576
5.9.2.	Organic acids	576
5.9.3.	Oil production	577
6.	Sugarcane bagasse: an ideal and model substrate for biorefinery approach.	577
7.	Conclusions and future prospects	577
	Declaration of competing interest.	577
	Acknowledgements	577
	Conflict of interest	577
	References	578

1. Introduction

Sugarcane is a major agricultural crop cultivated in tropical and sub-tropical countries in the world. India is the second largest country after Brazil for the production of sugarcane. Sugarcane belongs to Gramineae family and *Saccharum officinarum* is its botanical name. The estimated production of sugarcane crop in India is about 306 million tons (Mt), which is lower than Brazil (758 Mt) but higher than other countries [1,2]. Sugarcane is transported to mills after harvesting from fields and used for juice extraction used in sugar production. Nearly 80% of global sugar demand is fulfilled by sugarcane cultivation [3]. After industrial processing of sugarcane, major byproduct sugarcane bagasse is produced in large quantity [4]. It is usually generated after cleaning and extraction of juice from sugarcane. Sugarcane bagasse, a fibrous residue is consisting of nearly 32–45% cellulose, 20–32% hemicellulose, 17–32% lignin, 1.0–9.0% ash and some other components [5,6].

Due to excess production, bagasse is burnt as a mean of solid waste disposal which causes environmental pollution. Therefore, waste recycling has become the main issue of scientific research because of environmental concerns [7]. Supply of various agricultural residues usually varies according to weather, crop condition, and topography except some residues including sugarcane bagasse and rice straw [8]. Sugarcane bagasse can be harvested 100% from agricultural fields as compared to other residues (40%) as biofuel feedstock [8]. Utilization of this residue of sugar industry not only important in terms of environmental view but also acts as an effective substrate for the production of valuable and new products at large scale [9]. Sugarcane bagasse is also considered as a sustainable alternative biomass for the production of second generation biofuels (ethanol), electricity, enzymes, sugars and various other value-added products [10,11]. Bagasse is also used as biosorbent for removing the toxic and heavy metals like Pb, Zn, Cu, Cd and others [6]. Due to the presence of substantial cellulose and hemicellulose content, sugarcane bagasse has been used as a sustainable

biomass for biofuel production [12,13]. In the last few decades, overconsumption of non-renewable sources of energy i.e., fossil fuels has highlighted the environmental concern as well as energy crisis issues all over the world [14]. Severe climate problems related to greenhouse gas emissions, have compelled scientists to use alternative sources of energy in biorefinery [14].

Various bioconversion technologies for biofuel production have been reported for >5 decades, but now a days, biorefinery approach to convert numerous lignocellulosic feedstock into valuable products by various process technologies, with main focus on various biofuels and byproducts related to biochemical and bioenergy highly gaining the researchers interest [15]. Biorefinery approach offers malleability of various feedstocks and promising energy efficiency related to biofuel production [16]. Lignocellulosic derived bioethanol usually comprises of three steps including pretreatment, saccharification and fermentation. Although first generation biofuels reduce the CO₂ emission in comparison to fossil fuels but also demands high production of feedstock, developing competition in food crops that could enhance the food prices i.e., limits the economic growth. In spite of this, second generation bioethanol can solve these problems as they utilizes easily available, cost effective, non-food resources i.e., lignocellulosic materials [14].

The structural sugars of bagasse related to cellulose and hemicellulose, make it a potential substrate for production of biofuels, organic acids, enzymes and other value-added products in sugarcane mills. But presence of lignin hampers the saccharification of sugarcane bagasse process due to non-productive binding of the enzymes [10,17]. Actually lignin component of bagasse acts as a major barrier for effective conversion of cellulose and hemicellulose into various monomeric sugars for biofuels production [10,18]. Therefore, there is requirement of an appropriate pretreatment strategy that effectively solubilizes the hemicellulose as well as exposes the cellulose for cellulolytic enzymes. Pretreatment also decreases the particle size, polymerization and crystallinity of cellulose, resulting in increased surface area and porosity

that promotes enzymatic digestability of biomass [19,20]. Pretreatment is followed by a saccharification that converts lignocellulosics into high sugar yields [21,22]. The next step in the production of lignocellulosic biofuels is fermentation in which sugars (pentoses and hexoses) of enzymatic hydrolysate are utilized by various microorganisms including bacteria and yeasts for the production of biofuels and organic acids.

Most commonly used processes for fermenting sugars into biofuels are simultaneous saccharification and fermentation (SSF) and separated hydrolysis and fermentation (SHF). Sugarcane bagasse is directly or indirectly has been used in the generation of electricity, syngas and biogas [23]. Bagasse is used as feed for cattles due to the presence of nutrients. Sugarcane bagasse has been utilized for the bioremediation of heavy

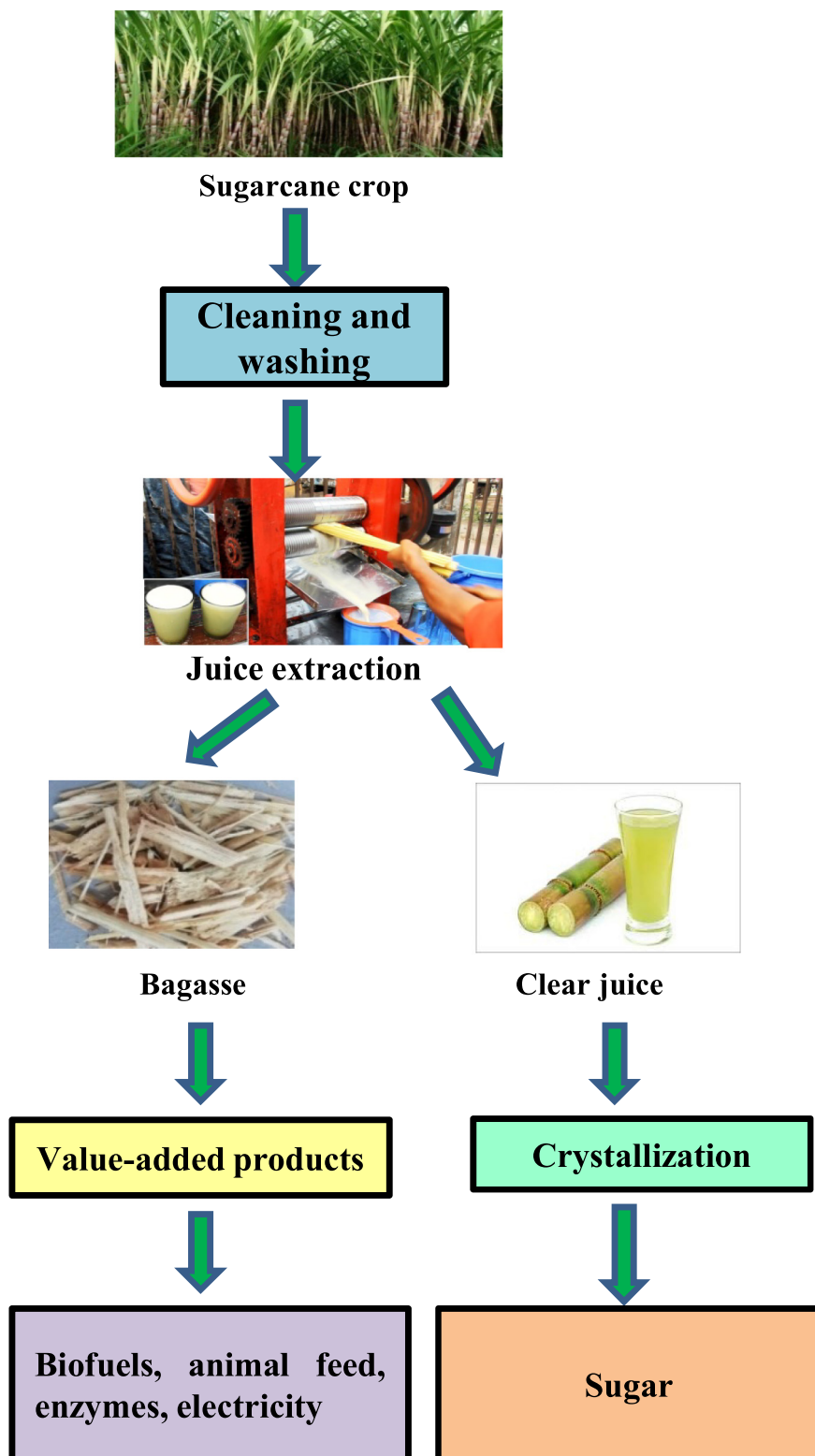


Fig. 1. Simplified process diagram for the generation of sugarcane bagasse.

metals from waste water [24,25]. Due to the presence of various binding sites on the surface of bagasse, it is used as biosorbent for the removal of toxic heavy metals from the environment. This review describes the sustainable and efficient utilization of sugarcane bagasse for the cleaner production of various value-added products.

2. World scenario for occurrence of sugarcane bagasse

Sugarcane is an economically important crop grown all over the world. India is the 2nd largest country for the production of sugarcane with an estimated production of around 306 million tons [26]. Sugarcane is an indispensable crop used for the production of sugar (sucrose) and ethanol. After being mechanically harvested, it is transported to mill for the extraction of raw juice, from which major byproduct obtained is sugarcane bagasse (Fig. 1). This byproduct bagasse is also produced in large scale i.e., 540 million metric tons per year basis globally among the top ten countries [27]. Therefore, sugarcane mills incurred additional disposal cost for bagasse. Also, the sugarcane mills management faces problems regarding regulations of cleaning air from the EPA because most of the bagasse is burnt in open which causes air pollution. Therefore, there is need of utilizing this excess of bagasse for the production of various value-added substances with great biotechnological applications [28].

3. Composition, mechanical properties and structure of sugarcane bagasse

3.1. Composition

Sugarcane bagasse is a fibrous residue consisting of cellulose (32–45%), hemicelluloses (20–32%), lignin (17–32%), ash (1.0–9.0%) and some extractives [5,6,29]. Due to low ash content, bagasse has various advantages in microbial bioconversion as compared to other biomass like wheat straw and rice straw having 11.0 and 17.5% ash contents, respectively. Due to high regeneration capacity and yield, bagasse can also be considered as a rich solar reservoir i.e., 80 t/ha in comparison to about 1, 2 and 20 t/ha for wheat, other grasses and trees, respectively [30].

3.2. Mechanical properties

Main mechanical properties of sugarcane bagasse include flexural strength and modulus, tensile strength and modulus, weathering, durability, impact strength and flame retarding. Flexural strength is influenced by various factors like fibre length, rind, powder's size and content. The flexural strength of bagasse first increases by increasing bagasse content followed by decline on further addition of bagasse [31,32]. In contrast, one study showed stability in flexural strength due to addition of bagasse [33]. According to some reports, tensile strength of bagasse also increases with addition of bagasse content [34,35]. While some literature suggested that it decreases with the addition of bagasse content and decreases faster when the bagasse powder is coarser [36,37]. However, tensile modulus generally increases with bagasse content [37,38]. Impact strength of bagasse decreases with increasing bagasse content [36] while Chaharmahali et al. [34] reported increase in tensile strength with increase in bagasse content. There is slower reduction and higher impact strength in case of finer bagasse [32,36]. Bagasse fibre is susceptible to microbial degradation like fungal attack, which causes weight loss. Literature survey showed that *Coniophora puteana* (brown rot fungus) resulted in more weight loss as compared to *Trametes versicolor* (white rot fungus) due to release of oxalic acid by the former [39]. Brown rot fungus hydrolyzes hemicellulose by producing oxalic acid making cellulose available for enzymatic hydrolysis and leaving only lignin skeleton. While white rot fungus not only decreases the degree of polymerization but also attacks surface of microfibrils [40,41]. Flexural modulus and strength is decreased by fungal

attack but hardness and impact strength reduction is very low [39,42]. Due to degradation of lignocellulosic material by fungus, more pores are formed which increases water absorption. Initially moisture content increases that decreases with extending fungal exposure time [42]. Antifungal agents (zinc borate) can be added to suppress fungal decay, which is also eco-friendly with low volatility and mammalian toxicity [43,44].

Weathering is also responsible for changes in colour, chemical properties, wettability, visual appearance and mechanical properties. Lignin index, wood index, carbonyl index, and hydroxyl index decreases due to weathering as observed by FTIR analysis. In contrast, non-pigmented samples showed different results like decrease in hydroxyl and wood index while carbonyl and lignin index showed less decrease. This may be due to weaker bonds in pigmented samples which cause higher surface fibre removal [45].

3.3. Structure

Due to the presence of complex carbohydrate biopolymers in bagasse, there are four bonds present between the monomer units like ether bonds, ester bonds, hydrogen bonds and carbon-carbon bonds [46]. Ether bonds are present between the functional groups, ester bonds between the polysaccharides (hemicelluloses and cellulose) and lignin, hydrogen bonds between carbon-carbon in aromatic rings and the cellulose polymer chains. Cellulose fractions form the primary structural component, with a long linear homopolysaccharide of anhydroglucose joined by β -1,4 glycosidic bonds. Due to extensive inter- and intra-molecular hydrogen bonds, crystalline structure forms that eases its aggregation into cellulose fibrils. The molecular mass of bagasse cellulose ranges from 157,800 to 168,400 g/mol and its cellulose fibres size range from 1.0 to 1.5 mm. Microfibrils formed due to the assembly of cellulose units and further macrofibrils formed by its cross-linkage with hemicellulose matrices, this creates resistance to enzymatic and chemical hydrolysis. Number of glucose units in the polymer is related to its degree of polymerization and it also affects enzymatic hydrolysis. Cellulose I is the main polymorph of cellulose and can be detected through XRD. Crystallinity index is also detected through the technique XRD and it was found that the CrI of bagasse is 56.7% [14].

Hemicellulose, a heteropolysaccharide in bagasse is comprised of β -1,4 xylopyranose backbone, β glucans, xyloglucans, glucomannans, galactomannans and some amounts of uronic acids. Its average amount in bagasse is 7380 g/mol. Ratio of arabinose/xylose affects the degree of branching in hemicellulose means higher the ratio, lower the degree of polymerization and vice versa. Lignin is a complex polymer which combines the hemicellulose and cellulose. The functional groups which related to the lignin are hydroxyl group, glycosidic linkage, ether linkage and aromatic ring [14,47]. Lignin is the most abundant aromatic amorphous three dimensional phenolic biopolymer. According to some literature study, lignin derivatives from non-woody biomass are composed up of three main monolignols units like syringyl subunit (S), guaiacyl subunit (G) and *p*-hydroxyphenyl subunit. Lignin in sugarcane bagasse has average molecular weight of 507 to 3973 mol/g [48–50]. Because of the presence of G and S lignin in addition of H, lignin in bagasse is different from hardwood [51]. In addition of this, lignin fractions also contain syringic acid, ferulic acid, vanillic acid, *p*-coumaric acid, xylose, glucose, arabinose, galactose, acetosyringone and syringaldehyde [6,48,49]. Sugarcane bagasse is also an important source for the production of chemical compounds like lactic acid, glucose, furfural, arabinose, xylose and some polymeric materials [52–54].

4. Pretreatment strategies for sugarcane bagasse

Pretreatment has been used to enhance the accessibility and digestibility of cellulose for bioethanol production. Usually pretreatment with efficient carbohydrates and lignin recovery has been favoured

nowadays to reduce the biomass recalcitrance i.e., cellulose crystallinity, structural complexity of cell wall and lignification for its effective utilization in biorefinery. Various pretreatment strategies (Fig. 2) have been reported for sugarcane bagasse [55,56]. A single pretreatment strategy cannot be applicable for all lignocellulosic biomass in all conditions. Therefore, some important criteria related to maximum recovery of important polymers and compounds of biomass should be considered during pretreatment. Further, the severe pretreatment conditions degrade the hemicellulosic sugars that resulted in the formation of inhibitory compounds such as furfural and organic acids. Therefore, the formation of inhibitors during the pretreatment must be minimized as they negatively affect the enzymatic hydrolysis and fermentation [57]. Therefore, all these factors should be considered before selecting an appropriate pretreatment for lignocellulosic biomass.

4.1. Physical pretreatments

4.1.1. Mechanical pretreatment

Mechanical pretreatment reduces the particle size of biomass to make the material handling easy by increasing the surface to volume ratio, resulting in enhanced enzymatic accessibility. Mechanical pretreatment plays an important role in enhancing the carbohydrate output, development of significant particles size, enzymatic digestibility and bioconversion efficiency without toxin formation. Mechanical fractionation is an essential requirement for the separation of sugarcane bagasse into its respective fractions. It usually reduces the particle size that affects the molecular structure of biomass to stimulate the enzymatic accessibility as well as increases the affinity between enzyme and cellulose to enhance the hydrolysis rate [58]. This method seems to be an attractive pretreatment to enhance the hydrolysis yield from various lignocellulosic biomass. Usually small sized particles can readily be hydrolyzed in comparison to large ones as they have high specific surface area for enzyme and substrate interaction for efficient bioconversion [58]. Mechanical activation modifies the structure as well as physicochemical properties of bagasse by disrupting the chemical bonds, resulting in reduction of particle size. Thus, this pretreatment proved

to be an important step in transforming the feedstock to various energy biomaterials in various applications [59]. In spite of this, the complex matrix of bagasse demands the assistance of additional pretreatments (chemical, biological or physiochemical) [60]. Therefore, the method is considered as an efficient environment friendly physical pretreatment to reduce the recalcitrancy of bagasse as a result of modification in lignin structure [61,62].

4.1.2. Ultrasound pretreatment

Ultrasound pretreatment involves the disruption of cell wall structure to increase the available surface area of biomass for the formation of reactive cellulosic fraction for fermentable sugars [63]. This pretreatment generates cavitation bubbles in aqueous media that promotes the dissociation of water molecules into free radicals, resulting in dissociating the lignin and xylan linkage [64]. This pretreatment exhibits both mechanical as well as chemical effects on bagasse, exclusively at low frequency. The mechanical properties of ultrasound waves effectively degrade the organic framework of sugarcane bagasse. Major advantage of this pretreatment includes significant extraction of cellulose, hemicellulose and lignin, thus improving the enzymatic hydrolysis [65]. Velmurugan and Muthukumar [66] performed ultrasound-assisted alkaline pretreatment of sugarcane bagasse at 24 kHz, 400 W, 30–50 °C for 5–50 min, and achieved 99.6% delignification of biomass. Similarly, Esfahani and Azin [65] made a slurry of bagasse (10%) followed by addition of sulfuric acid and exposed the slurry to ultrasound pretreatment at 120 W for 180 s that resulted in efficient delignification.

4.1.3. Microwave pretreatment

Microwave pretreatment is low energy utilizing, convenient pretreatment method that dissociates the structural framework of cellulosic bundles with minimum inhibitors formation [67]. It has been considered as a non-conventional irradiation method that utilizes electromagnetic radiation to dissociate the structural make up of sugarcane bagasse. This pretreatment plays an important role in uniform distribution of heat in biomass whereas conventional heating methods result in generation of temperature gradients. Dielectric polarization of this

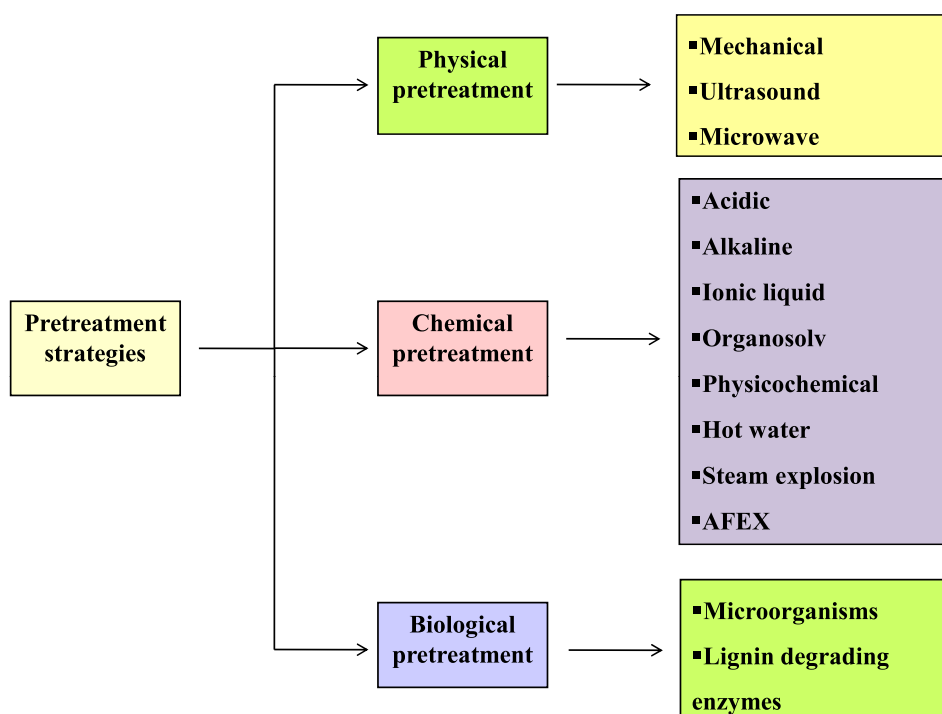


Fig. 2. Various physical, chemical and biological strategies used for the pretreatment of sugarcane bagasse.

pretreatment generates molecular collisions that dissociates the complex structure of sugarcane bagasse with high energy efficiency [68,69]. Microwave pretreatment is usually applied with other pretreatment methods for better results. de Souza Moretti et al. [70] carried out microwave pretreatment of sugarcane bagasse (1–3 mm) by immersing in distilled water and exposed the slurry at 2450 MHz for 5 min. Similarly, a study on microwave-assisted sulfolane pretreatment of sugarcane bagasse (10%) at different temperatures and reaction time was investigated by Portero-Barahona et al. [69].

4.2. Chemical pretreatments

4.2.1. Acid pretreatment

Acid pretreatment is an effective pretreatment approach that breaks down hemicellulose effectively and promotes the liberation of sugars from bagasse. This pretreatment has been usually adopted by researchers at commercial scale. This method includes dilute (<10%) and concentrated (>10%) acids for the pretreatment of sugarcane bagasse [71]. Diluted acid pretreatment has been considered as a promising pretreatment technique at industrial scale that reduces the hemicellulose content and enhances the exposure of cellulose for enzymatic action [72,73]. Sugarcane bagasse is commonly pretreated by sulfuric acid but high corrosion and formation of degradation products are several disadvantages [74,75]. In view of this, some organic acids i.e., citric acid, acetic acid, tartaric and formic acid are the preferred alternatives for the pretreatment of sugarcane bagasse [76]. Dilute acid pretreatment has been preferred for efficient enzymatic hydrolysis as well as a suitable approach to form xylooligosaccharide (XOS) from sugarcane bagasse [77]. It usually involves organic or mineral acids, which have been reported previously to obtain XOS [78,79]. Mineral acids stimulate the production of xyloses and furfural that somewhat inhibits XOS yield [79] whereas, organic acid promotes the production of XOS with less furfural, low corrosion and less inhibitor formation [80]. Different organic acids i.e., oxalic acid, acetic acid and others have been considered as XOS producing reagents during acidic pretreatment [76]. Timung et al. [81] carried out acidic pretreatment of sugarcane bagasse with sulfuric acid (0.1–0.9 M) at a substrate to liquid ratio of 1:10 at 80–120 °C for 120 min. Yu et al. [82] also utilized acidic pretreatment of sugarcane bagasse using HCl (1.25%) at 130 °C for 10 min.

4.2.2. Alkaline pretreatment

Alkaline pretreatment has been considered as a cost-effective process with less formation of inhibitors [83]. But the difficulty in its recovery is the main disadvantage. Alkaline pretreatment is generally performed at lower temperatures and pressure that includes NaOH [84], Ca(OH)₂ [85] and ammonia [86] pretreatment for subsequently removal of lignin as well as different uronic acid related substitutions of hemicellulose present in sugarcane bagasse. Alkaline pretreatment is preferred for separating the cellulose with delignification in bagasse. Alkaline conditions usually swell the bagasse that decreases the crystallinity of cellulose. Yu et al. [82] carried out alkaline pretreatment of sugarcane bagasse using 0.18% NaOH at solid to liquid ratio of 1:6 at 110 °C for 1 h. Yu et al. [87] also carried out alkaline pretreatment of sugarcane bagasse using 1.0% NaOH at 120 °C for 10 min and observed effective saccharification and delignification. Similarly, Zhang et al. [88] utilized different loadings of alkaline hydrogen peroxide (1.25–6.25%) at temperature range of 60–160 °C to degrade the hemicellulose or/and lignin effectively to enhance enzymatic digestability.

4.2.3. Ionic liquids pretreatment

Ionic liquids are non-volatile solvents of high thermal stability. Ionic liquids reduce the crystallinity of cellulose along with removal of lignin up to 60% [89]. 1-Ethyl-3-methylimidazolium acetate ([Emim][Ac]) has been reported as an effective ionic solvent to dissolve the cellulose of bagasse [78]. Some other solvents used for pretreatment of bagasse include 1-butyl-3-methylimidazolium chloride ([Bmim][Cl]), 1-allyl-3-

methylimidazolium chloride ([Amim][Cl]), 1-butyl-3-methylimidazolium acetate ([Bmim][Ac]), and 1,3-dimethylimidazolium dimethyl phosphate ([Mmim][DMP]) [90,91]. Ionic liquids being the salts of cations and anions of various organic compounds, act as catalyst for removal of lignin from bagasse [60]. Bahrani et al. [91] used ionic liquid i.e., ([Mmim][DMP]) pretreatment of sugarcane bagasse at ratio of 1:20 at temperature range of 100–140 °C for 120 min as an effective pretreatment for bagasse. Yoon et al. [92] carried out ionic liquid pretreatment of sugarcane bagasse at 100 °C for different reaction time (0.5–8 h) with low energy requirement.

4.3. Physiochemical pretreatment

Physiochemical approaches include various physical parameters such as heat, compression, pressure, sonication along with chemicals (acids, alkali, organic solvents and liquid ammonia). This combinatorial approach alters the constituents of sugarcane bagasse, therefore enhances the enzymatic accessibility of bagasse. Different physiochemical approaches including different loadings and modifications in chemical and physical factors respectively, have been reported previously for the pretreatment of sugarcane bagasse [93,94]. Various physiochemical approaches act synergistically to dissolve the lignin for subsequent improving the enzymatic hydrolysis [95,96].

4.3.1. Steam explosion

Steam explosion is low cost and environment-benign pretreatment approach preferably considered in sugarcane industry [55]. It includes the exposure of bagasse to high pressured steam followed by explosive decompression of sugarcane bagasse resulting in hydrolysis and solubilization of hemicellulose and lignin, respectively [97]. It usually modifies the crystallinity of cellulose as well as degree of polymerization, hence promotes the enzymatic hydrolysis [98]. This pretreatment does not require chemicals, however an energy demanding approach. Silveira et al. [99] carried out steam explosion pretreatment of sugarcane bagasse with moisture of 50% (w/w) in a 65 L steam gun reactor at pilot scale for enhanced saccharification. Pitarello et al. [100] carried out phosphoric acid-catalyzed steam pretreatment at different temperature (180–220 °C) for different reaction time (5–7.5 min) and reported 180 °C for 5 min as optimum pretreatment conditions. Rabelo et al. [101] preferred H₂O₂ assisted steam pretreatment of sugarcane bagasse (at 25 °C for 60 min) for enhanced saccharification and bioethanol production.

4.3.2. Organosolv pretreatment

Organosolv pretreatment has been considered as an emerging strategy due to its ability of fractionating the bagasse into its constituents (cellulose, hemicellulose and lignin) in relatively pure form with effective recovery and recycling of respective solvent [102]. This pretreatment uses organic solvent with/without catalyst for pretreatment of sugarcane bagasse. Organic solvent allows recovery of most of the cellulose as the solids whereas most of the hemicellulose and lignin gets dissolves in these solvents. Therefore, reduction in recalcitrance with increased enzymatic accessibility of cellulose for bioethanol production can be achieved [103]. It generally uses low (methanol, ethanol) or high (glycol or glycerol) boiling point liquids in acidic or non-acidic conditions. Low boiling alcohols are commonly used organosolvents because of low cost and easily recovery during the process, but the requirement of special instrument for operating high pressure, makes the process costly [104]. In this context high boiling alcohols operating at low temperature and pressure and utilizing more energy for their recovery, are highly considered [102]. Since the 1970s, different organic solvents i.e., acetone, alcohols, chloroethanol, formaldehyde, phenol, amines and propionic acid with/without catalysts have been reported for pretreatment of various lignocellulosic biomass [105]. This pretreatment is mainly preferred for small mills due to its easily recovery benefit

with sulfur free discharge [56]. Several reported studies highlighted the efficacy of organosolvents for the conversion of sugarcane bagasse into cellulose and relatively pure lignin with proper recovery of value-added products that improves the economic viability of bioethanol production [106]. It usually utilized mixture of organic solvents such as ethanol and water, for pretreatment of bagasse for effective solubilization of lignin linkages [55]. Zhang and Wu [107] carried out pretreatment of bagasse with combination of ethanol and water in 60:40 ratio and 5% acetic acid at 190 °C for 45 min. Similarly, pretreatment of sugarcane bagasse with ethanol (60%) and FeCl_3 (0.025%) at 160 °C for 72 h has been reported by Zhang et al. [108].

4.3.3. Hot water pretreatment

Hot water pretreatment usually consumes high volume of water at high temperature in the range of 120–23 °C for 1–80 min, resulting in recovery of 55–84% hemicellulose [56,109]. It is an attractive pretreatment technique that uses ambient pressure at elevated temperatures i.e., 160–240 °C to keep the water in liquid state and attains theoretical enzymatic digestibility of cellulose without addition of any other chemical [110,111]. Hot water pretreatment has been considered as an eco-friendly process with reduced liberation of polysaccharides and less inhibitor generation [112]. Liquid hot water pretreatment also favours the solvation of hemicellulose with delignification in sugarcane bagasse [95,113]. Hongdan et al. [114] considered liquid hot water pretreatment of bagasse at solid to liquid ratio of 1:20 at 180 °C for 30 min and observed effective removal of hemicellulose with enhanced saccharification. Yu et al. [82] also reported delignification and better sugar recovery in liquid hot water pretreated sugarcane bagasse.

4.3.4. Ammonia fibre explosion

Ammonia fibre explosion (AFEX) is a preferred steam explosion type physiochemical pretreatment strategy, usually utilizes liquid ammonia for pretreatment at relatively high temperature and pressure followed by rapidly reducing the pressure after a period of time [115]. This process effectively dissolves the hemicellulose and reduces or depolymerizes the lignin, thus enhances the enzymatic hydrolysis. This pretreatment has been significantly utilized sugarcane bagasse for improving the delignification and saccharification rate [116,117]. Krishnan et al. [116] reported 85% glucan conversion in AFEX pretreated bagasse due to disruption of the ester and lignin-carbohydrates linkages. Similarly, pretreatment of sugarcane bagasse with ammonia (15%) at 170 °C for 60 min, resulted in enhanced delignification as well as saccharification [117]. Bala and Singh [118] also preferred ammonia pretreatment for enhanced saccharification of sugarcane bagasse.

4.4. Biological pretreatment

Biological pretreatment is an eco-friendly pretreatment approach that enhances the enzymatic accessibility of polysaccharides as a result of microbial or enzymatic delignification [119]. This pretreatment aims at selection of a suitable microorganism/enzyme to make the process successful. Now-a-days, this pretreatment has gained researchers interest due to enhanced recovery of biobased products with improved saccharification and fermentation [119]. This pretreatment is highly preferred over physical and chemical pretreatment in terms of less energy requirement, simple operating conditions with less inhibitors formation. Usually most of the fungi enhance the digestibility of polysaccharides but there are only a few species that also degrade lignin effectively with preserving the cellulose and hemicelluloses [120]. Previously reported studies related to biological pretreatment of sugarcane bagasse assured that selection of particular microorganism depends on type of biomass to be used in pretreatment [121]. White rot, soft rot and brown rot fungi are usually considered for degradation of hemicellulose and lignin of sugarcane bagasse [122]. However white rot fungi are the most preferred microorganism for the pretreatment of sugarcane bagasse for effective delignification by their lignolytic enzymes i.e. laccases

and peroxidases [123]. Lignolytic enzymes enhance the accessibility of cellulose as well as recover the important phenolic compounds formed after lignin degradation [119]. White rot fungi disintegrates lignocellulosics cell wall components with effective delignification. Biological pretreatment may be followed by effective saccharification for enhanced production of biofuels [123,124]. Khuang et al. [124] investigated the production of ethanol by integrated fungal fermentation (IFE) of bagasse using white-rot fungus *Phlebia* sp. MG-60. They evaluated the effect of various culture conditions including moisture content and various chemical factors (Mn^{2+} , Fe^{2+} or Cu^{2+}) on IFE efficiency. Jiraprasertwong et al. [125] used a white-rot fungus as an effective biological pretreatment for sugarcane bagasse. Machado and Ferraz [126] carried out biological pretreatment with *Ceriporiopsis submervispora* at 27 °C for 60 days for high sugar recovery. Laccase from *Pycnoporus cinnabarinus* was also used for effective delignification of sugarcane bagasse [127].

5. Value-added products from sugarcane bagasse

Sugarcane bagasse has been utilized tremendously for the production of various value-added products i.e., biofuels, organic acids (Table 1), enzymes (Table 2), metabolites, electricity generation, animal feed (Table 3) and many more [83,184] (Fig. 3). Previously reported studies related to biorefining of sugarcane bagasse, prove it as a relevant substrate for valuable products due to high content of polysaccharide. Sugarcane bagasse is well known lignocellulosic residue for production of second-generation biofuels also (Fig. 4) [185].

5.1. Biofuel production

5.1.1. Biohydrogen

Hydrogen is a carbon-free clean energy fuel that forms water as byproduct after combustion. Hydrogen has 3-fold high energy density in comparison to hydrocarbon fuels [186]. Hydrogen has gained researcher's interest due to its distinctive characteristics (colourless, odourless, less toxicity etc.) and has been considered to play a key role in future energy requirements [187]. Now a days, hydrogen production from abundant and renewable agriculture residues are usually preferred. Jiang et al. [143] pretreated the sugarcane bagasse by two-step dilute acid and observed 2.06 mol H_2 /mol of total reducing sugars in pretreated hydrolysate (15.66 g/l glucose). Similarly, Ratti et al. [142] used steam explosion and alkaline delignification pretreated sugarcane bagasse and reported 1.2 mol H_2 /g substrate by thermophilic microbial consortium using sugarcane bagasse at 55 °C in a batch reactor. Usually steam explosion strategy hydrolyzes the sugarcane bagasse in two fractions to enhance the accessibility of enzymatic hydrolysis (a) aqueous fraction that contains mainly xylose and other hemicellulosic sugars (b) solid fraction that contains glucose rich cellulose and lignin. Whereas, alkaline pretreatment catalyzes the removal of lignin from solid fraction [142].

5.1.2. Bioethanol

Currently, ethanol is the most favourable biofuel in comparison to others as it has low CO_2 emission, high energy density and heat of vapourization. It is an alcoholic beverage preferred as industrial ingredient as well as a base chemical in the formation of various organic compounds such as acetic acid, ethyl esters and diethyl ether [188]. Lignocellulosic derived ethanol is considered to be an alternative of pollution causing fossil fuels. Sugarcane bagasse due to its huge availability and high cellulose content, has attracted researchers for bioethanol production since last decades [189]. Secondly, it has less carbon intensive properties in comparison to fossil fuels that reduces air pollution and makes it a preferred source for the production of ethanol [189,190]. Several parameters i.e. quality of bagasse, employed process for bioethanol production highly affect the ethanol yield [191]. According to some reports, theoretical yield varied from 157 to 225 L of ethanol/ton of

Table 1
Production of biofuels and organic acids from sugarcane bagasse using microorganisms.

Microorganism	Process used	Product	Yield (g/l)	Reference
<i>Zymomonas mobilis</i>	Acid pretreatment [1% (v/v) H ₂ SO ₄ at 121 °C for 30 min]	Bioethanol	60	[128]
<i>Klebsiella pneumoniae</i> CGMCC 1.9131	Alkali-PAA pretreatment was performed using 10% NaOH (3:1 liquor to solid ratio) for 1.5 h at 90 °C followed by 15% PAA treatment at 75 °C for 3 h	Bioethanol	50.6	[129]
<i>Saccharomyces cerevisiae</i>	Steam explosion pretreatment at 215 °C for 5 min	Bioethanol	56.3	[130]
<i>Saccharomyces cerevisiae</i>	Delignification by 1NaOH (1 N) for 2 h	Bioethanol	11.81	[131]
<i>Pichia stipitis</i>	Acid pretreatment was carried out using 4% (v/v) H ₂ SO ₄ at 121 °C for 90 min	Bioethanol	3.70	[132]
<i>Saccharomyces cerevisiae</i>	Salt-alkali pretreatment carried out, first by ZnCl ₂ (1.73 M) followed by NaOH (1.36 M) and both at 121 °C for 30 min	Bioethanol	4.88	[133]
<i>Saccharomyces cerevisiae</i>	Acid alkaline pretreatment carried out, 1st by H ₂ SO ₄ (2% v/v) pretreatment followed by NaOH (4% w/w), both at 121 °C, 30 min	Bioethanol	29.83	[134]
<i>Saccharomyces cerevisiae</i> , <i>Pichia stipitis</i>	Fermentation of ammonia pretreated sugarcane bagasse hydrolysate (20%) by yeasts individually as well as combination at 30 °C, 72 h and 150 rpm	Bioethanol	31.56	[118]
<i>Candida tropicalis</i>	Ammonium hydroxide (20%) pretreatment at 50 °C for 48 h	Bioethanol	57.2	[135]
<i>Scenedesmus acutus</i> PPNK1	Hydrolysis with 0.5% (v/v) H ₂ SO ₄ and deionized water under autoclave for 15 min at 121 °C	Biodiesel	0.160	[136]
<i>Klebsiella pneumoniae</i>	Alkali-PAA pretreatment was performed using 10% NaOH (3:1 liquor to solid ratio) for 1.5 h at 90 °C followed by 15% PAA treatment at 75 °C for 3 h	2,3-Butanediol	17.35	[137]
<i>Clostridium beijerinckii</i> NCIMB 8052	Sequential pretreatments were carried out like milling, hot water pretreatment at 200 °C for 1 h, microwave pretreatment for 7 min at 120 °C than ammonia immersion (2:1; v/w) for 30 min at 90 °C and at last enzymatic hydrolysis	Biobutanol	6.86	[138]
<i>Clostridium acetobutylicum</i>	Combined dilute acid with oxidate ammonolysis pretreatment was performed by 1% H ₂ SO ₄ for 1 h at 140 °C than washed, dried bagasse pretreated using 5% NH ₃ H ₂ O with 6% H ₂ O ₂ for 6 h at 80 °C	Biobutanol	4.62	[139]
<i>Clostridium butyricum</i> TISTR 1032	Immobilization was done by using delignified bagasse (7 g) in 63 ml of tryptone sucrose yeast extract which was autoclaved for 15 min at 121 °C before addition of 10% inocula of <i>Clostridium butyricum</i>	Biohydrogen	1.52 mol H ₂ /mol hexose consumed	[140]
<i>Thermoanaerobacterium aotearoense</i>	H ₂ SO ₄ (2.3%) pretreatment at 115 °C for 114.2 min	Biohydrogen	520	[141]
Thermophilic microbial consortium	Steam explosion for 7 min at 200 °C followed by alkaline delignification pretreatment with NaOH for 30 min at 121 °C	Biohydrogen	1.2 mol H ₂ /g substrate hydrogen	[142]
<i>Clostridium butyrium</i>	Two-step dilute acid hydrolysis with H ₂ SO ₄ solution (2 wt%) with solid-liquid ratio (SRL) of 1:10 (g/g) for 1 h at 130 °C followed by H ₂ SO ₄ solution (4 wt%) with SRL of 1:10 (g/g) for 1 h at 150 °C	Biohydrogen	15.66	[143]
<i>Debaromyces hansenii</i>	Steam exploded bagasse was hydrolyzed with H ₂ SO ₄ (0.1%) for 30 min at 100 °C followed by detoxify with activated charcoal (1%) for 4 h at 28 °C, 50 rpm	Xylitol	73.8	[144]
<i>Williopsis saturnus</i>	Dilute H ₂ SO ₄ pretreatment at 1:10 solid/liquid ratio for 20 min at 121 °C followed by treating with addition of activated charcoal (1%, w/v)	Xylitol	0.51 g/g xylose consumed	[145]
<i>Candida tropicalis</i> IEC5-ITV and <i>Saccharomyces cerevisiae</i>	Acidic and enzymatic hydrolysis	Ethanol and xylitol	Ethanol (19.53), xylitol (10)	[146]
<i>Candida tropicalis</i>	Mixed alkali solution (NaOH/ammonia ratio of 2:1) followed by ultrasonic treatment at 400 W, 45 °C for 40 min	Xylitol	62.9800	[147]
<i>Candida tropicalis</i> JA2	Steam-explosion for 8 min at 15 kgf/cm ² , 15% humidity followed by H ₂ SO ₄ (0.5%, w/w) for 100 min at 130 °C	Xylitol	109.5	[148]
<i>Candida tropicalis</i>	Ammonium hydroxide (20%) pretreatment at 50 °C for 48 h	Xylitol	34.5	[135]
<i>Aspergillus niger</i> DS1	–	Citric acid	6.2 g/100 g of dry solid	[149]
<i>Aspergillus niger</i>	–	Citric acid	18.63	[150]
<i>Actinobacillus succinogenes</i>	Acid pretreatment with H ₂ SO ₄ (1%, v/v) at 121 °C for 40 min	Succinic acid	22.5	[151]
<i>Aspergillus niger</i>	–	Itaconic acid	8.241 mg/kg	[152]
<i>Bacillus</i> sp.	Acid-base-enzyme joint treatment carried out by first with 5 L H ₂ SO ₄ (1.5%) at 121 °C for 1 h then with 5 L NaOH (1.5%) at 100 °C for 1 h and at last cellulose hydrolysis	Lactic acid	185	[153]
<i>Clostridium tyrobutyricum</i>	Dilute acid pretreatment with dilute HCl solution (1 ml, 0.1–0.5 N) at autoclaved time of 30 min, 121 °C	Butyric acid	20.9	[154]
Immobilized <i>Propionibacterium freudenreichii</i>	Bagasse was chopped using a crushing mill into small pieces (1–10 mm) followed by autoclaving at 121 °C for 20 min	Propionic acid	136	[155]
Immobilized <i>Propionibacterium acidipropionici</i>	Sugarcane bagasse was hydrolyzed with 0.3 M H ₂ SO ₄ at a ratio of 10:1 at 121 °C for 30 min	Propionic acid	51.2	[156]

bagasse [192]. Bala and Singh [118] carried out the fermentation of ammonia pretreated sugarcane bagasse hydrolysate (20%) by *Saccharomyces cerevisiae* and *Pichia stipitis* individually as well as combination at 30 °C and 150 rpm that resulted in 24.56, 23.56 and 31.56 g/L bioethanol, respectively after 72 h. Pretreatment of sugarcane bagasse pith with sulfuric acid resulted in 3.70 g/L ethanol in simultaneous saccharification and fermentation [132]. Wahono et al. [131] reported maximum ethanol of 11.81 g/L by *S. cerevisiae* from sugarcane bagasse hydrolysate. Amores et al. [130] found 56.3 g/L ethanol from

hydrolysate of steam explosion (215 °C for 5 min) pretreated sugarcane bagasse in simultaneous saccharification and fermentation.

5.1.3. Biobutanol

Biobutanol is also known as next generation liquid biofuel usually formed by fermentative microbes in acetone-butanol-ethanol (ABE) fermentation. Butanol has various properties similar to gasoline, therefore, also considered as a dye solvent and preferred fuel in industries. It has high energy in comparison to ethanol, so preferring in transport

Table 2
Production of different enzymes by microorganisms using sugarcane bagasse.

Microorganism	Process used	Product	Yield (U/g or U/ml) ^a	Reference
<i>Trichoderma auroviride</i>	Alkaline hydrolysis was carried out by using NaOH (1 L, 0.25 N) at 121 °C for 4 h	Xylanase	2037 U/g	[157]
<i>Trichoderma</i> sp.	Milling and then fungal hydrolysis at 30 °C	Xylanase	380 U/g	[158]
<i>Trichoderma viride</i> -IR05	KOH (2.5%) pretreatment at ratio of (1:10, solid:liquid) at room temperature for 2 h	Xylanase	72.4 U/g	[159]
<i>Aspergillus niger</i>	Fungal hydrolysis at 28 °C, 5d	Xylanase	3400 U/ml	[160]
<i>Bacillus subtilis</i> BS04	Bacterial hydrolysis at 35 °C for 48 h	Xylanase	47.33 U/ml	[161]
<i>Aureobasidium pullulans</i>	Fungal hydrolysis at 30 °C	Xylanase	0.284 U/ml	[162]
<i>Bacillus subtilis</i> subsp. <i>subtilis</i> JJBS250	Fine bagasse (5 g) with moistening medium (20 ml) after autoclaving, incubated at 40 °C for 48 h	Xylanase	98.16 U/g	[163]
<i>Trichoderma harzianum</i> L04	Acid and alkali pretreatment, first by H ₂ SO ₄ (3%) then by NaOH (4%), both for 20 min at 121 °C	Cellulase	1.968 U/ml	[164]
<i>Penicillium echinulatum</i>	Steam pretreatment at 200 °C, 7 min	Cellulase	2.6 FPU/ml	[165]
<i>Aspergillus flavus</i>	Fungal hydrolysis at 28 °C for 69 days	Cellulase	33.15 U/g	[166]
<i>Trichoderma koningii</i>	Acid and alkali pretreatment was performed in sequence; first by H ₂ SO ₄ (1%) followed by NaOH (7%) both for 40 min at 100 °C	Cellulase	8.2 U/g	[167]
<i>Bacillus subtilis</i> KCC 103	Dilute H ₂ SO ₄ pretreatment autoclaved for 30 min	α-Amylase	144.8 U/ml	[168]
<i>Bacillus licheniformis</i>	Bacterial hydrolysis at 33 °C for 120 h	Protease	146.28 U/g	[169]
<i>Rhizopus oryzae</i>	Fungal hydrolysis for 48 h at 30 °C	Lipase	215 U/g	[170]
<i>Aspergillus fumigates</i>	Fungal hydrolysis for 24 h at 32 °C		40 U/ml	[171]

^a U/g for solid state fermentation (SSF) and U/ml for submerged fermentation (SmF).

industry now a days [188]. Butanol is usually formed by *Clostridium* spp. such as *C. acetobutylicum*, *C. saccharobutylicum* and *C. beijerinckii*. Currently efforts for lignocellulosic derived butanol have been considered at industrial scale [193]. Recently, various studies reported the high potential of sugarcane bagasse to produce butanol [138,194]. Su et al. [138] reported 6.86 g/L butanol using combinatorial pretreatment (milling, liquid hot water and microwave) by *Clostridium beijerinckii* NCIMB

8052 in simultaneous saccharification and fermentation. Similarly, oxidative ammonolysis pretreated sugarcane bagasse resulted in 4.62 g/L butanol using *C. acetobutylicum* ATCC 824 [139].

5.1.4. Xylitol

Xylitol is a natural and five carbon alcohol found in many fruits and vegetables. Xylitol is a highly preferred sugar substitute in food and

Table 3
Production of other value-added products from sugarcane bagasse.

Microorganism	Process used	Product	Effect on product	Reference
<i>Pleurotus eryngii</i>	Substrate was incubated at 24 °C for 35 days with spawn of <i>Pleurotus eryngii</i> under semi-anaerobic condition	Animal feed	Increase digestibility of bagasse by <i>Pleurotus eryngii</i>	[172]
<i>Trichoderma harzianum</i>	Chopped bagasse was ammoniated with urea (4%) and water (1:1) than incubated for 21 days and at last fermented with <i>T. harzianum</i> (10%)	Animal feed	Nutritional value of sugarcane bagasse has been improved after ammonication of sugarcane bagasse	[173]
–	Urea (5%), molasses (5%) fermentation of bagasse for 21 days	Animal feed	After urea, molasses treatment of sugarcane bagasse, nutritional value increased which is used as suitable feed for dairy cattles	[174]
–	Citric acid (8.3 ml solution/g of bagasse ratio) was added to bagasse at 30 min of agitation	Bioadsorbent	Increased adsorption capacity of citric acid modified sugarcane bagasse for Cu(II) as compared to raw bagasse	[175]
–	Phthalic anhydride modification of bagasse for 3 h at 100 °C, 300 rpm in oil bath placed heating plate	Bioadsorbent	Increased capacity of carboxylate functionalized sugarcane bagasse in removing nickel ions	[24]
–	Hydrolysis of bagasse with mixture of glacial acetic acid (8.75 M) and hydrogen peroxide (21.6 M) at ratio of 1:1 for 48 h at 60 °C	Xylooligosaccharides	Combinatorial pretreatment of bagasse with acetic acid, hydrogen peroxide followed by the recombinant xylanase from <i>Bacillus subtilis</i> for more xylooligosaccharide production	[176]
<i>Clostridium thermocellum</i>	Hydrolysis of bagasse with XynZ from <i>C. thermocellum</i>	Antioxidant compounds and xylooligosaccharides	Simultaneous production of antioxidant compounds and xylooligosaccharides from enzymatic hydrolysate of sugarcane bagasse	[177]
<i>Gluconobacter oxydans</i>	Prehydrolysis of bagasse with acetic acid (10%) for 45 min at 150 °C	Xylooligosaccharides	Maximum xylooligosaccharides production from this hydrolysis	[178]
–	Dil. H ₂ SO ₄ (0.5% w/v) or ([Emim][Ac]) pretreatments for 30 min at 100 °C	Xylooligosaccharides	This pretreatments effects better xylooligosaccharides production	[179]
<i>Candida langeronii</i>	Sulfuric acid (70 mg/g of dry mass) pretreatment for 2 h at 125 °C followed by hydrolysis by <i>Candida langeronii</i>	Single cell protein	Utilized pretreated sugarcane bagasse for effective production of single cell protein by <i>Candida langeronii</i> RLJY-019	[180]
<i>Saccharomyces cerevisiae</i>	NaOH (2%) pretreatment for 2 h at room temperature followed by fermentation by <i>S. cerevisiae</i> for 72 h	Single cell protein	Utilization of good amount of SCP from NaOH pretreated sugarcane bagasse	[181]
<i>Pycnoporus sanguineus</i> MCA 16	Hydrothermal alkaline pretreatment (0.5 mol/l NaOH solution) followed by enzymatic saccharification	Reducing sugars	Increase in reducing sugar production	[182]
–	First delignification using NaOH (15%) and Na ₂ S (5%) for 2 h at 120 °C followed by gamma irradiation (485 kCi) and at last H ₂ SO ₄ hydrolysis (0.05%) at 121 °C for 2 h Pretreatment was carried out by Na ₂ CO ₃ (1% w/v) for 60 min, 121 °C	Reducing sugar	Three-fold increase in reducing sugar production	[183]
<i>Bacillus subtilis</i> subsp. <i>subtilis</i> JJBS250		Reducing sugar	Untreated bagasse liberated high amount of reducing sugar as compared to pretreated	[163]

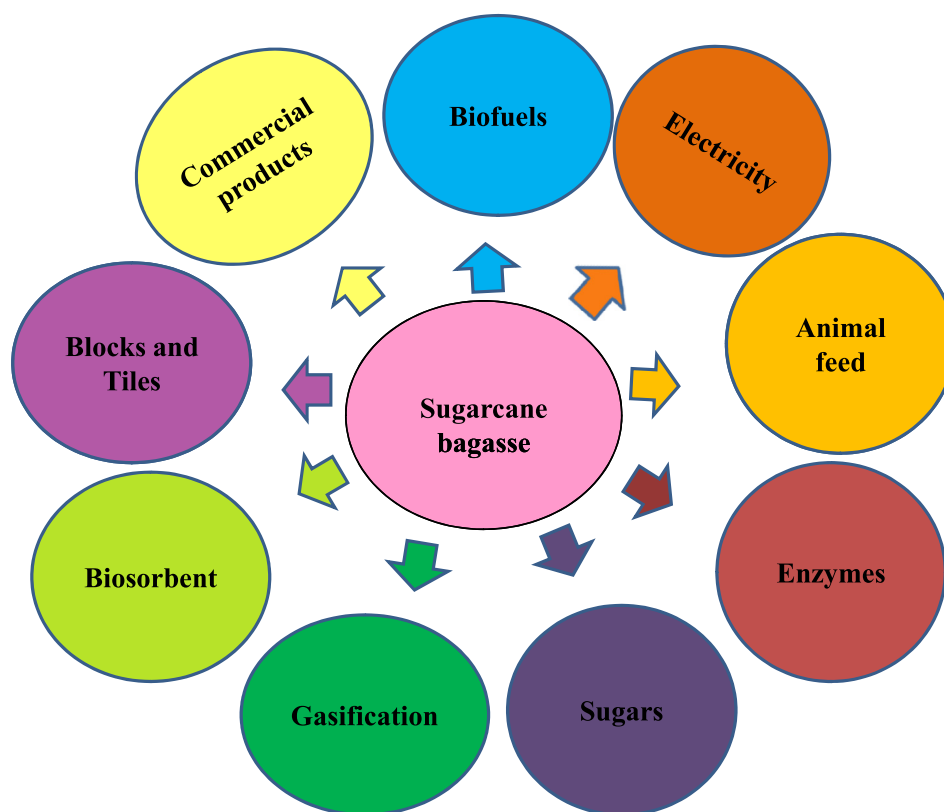


Fig. 3. Different value-added products generated from sugarcane bagasse.

pharmaceutical industries due to its remineralization and anti-carcinogenic properties [148,195]. It has been utilized as a sugar substitute in dietary foods related to diabetic patients [147,195]. Production of xylitol from xylose by microbial fermentation has been gaining importance as it utilizes waste for conversion into valuable products, hence plays an important role in resources conservation as well as environment protection [147]. Xylose metabolism usually depends on the type of microbes (bacteria, filamentous fungi and yeasts) used in the xylitol fermentation. Among > 100 types of microbes, *Candida* sp. has been utilized for xylitol production due to high xylose conversion efficiency and high tolerance ability to various inhibitors (furfural, phenolics and acetate) present in xylose liquor [147]. Optimization of various variables (solid to liquid ratio, temperature, time and pressure) were found to enhance the hemicellulose (xylose) recovery from bagasse. Among various lignocellulosic residues, sugarcane bagasse has high potential as a suitable raw material for conversion of xylose into xylitol. Various lignocellulosic materials including sugarcane bagasse have been reported as a suitable raw material for the production of xylitol. Sugarcane bagasse was used as entrapping carrier for *Candida guilliermondii* cells for xylitol production [196]. However, there is need of more efforts for xylitol production at industrial scale. According to report of 2013, estimated market value of xylitol is about US\$670 million, which is supposed to increased up to US\$1 billion by 2020 [2]. Xu et al. [147] found 62.98 g/L xylitol using 100 g/L xylose by *C. tropicalis* 31949 using sugarcane bagasse as substrate after 26 h. *Candida tropicalis* JA2 produced 109.5 g/L of xylitol using sugarcane bagasse hydrolysate [148]. Raj and Krishnan [135] reported 34.5 g/L of xylitol from ammonium hydroxide pretreated sugarcane bagasse.

5.2. Electricity generation

For the foreseeable future, countries are searching for the renewable energy resources for the electricity generation. Sugarcane bagasse is one of the widely used lignocellulosic biomass as compared to other

biomass residues for the energy generation [197,198]. Both in environmental and economic terms, the use of sugarcane bagasse as a fuel is a good initiative for supplying electricity to the national grid [199]. The energy production from bagasse is very economical and renewable as it supplies the entire energy requirements of the sugar and ethanol production processes as well as produces surplus energy for industry itself and also sold to the grid for reducing the cost of electric power [200].

Presently, new mill projects contribute in cogeneration plants to generate surplus electricity. Cogeneration is a process of producing thermal energy and electricity at high energy output. To generate higher amount of bioenergy from bagasse, more efficient cogeneration technologies have been used [201]. Three main processes used to convert bagasse into energy are physical, biological and thermochemical. Among these, direct combustion is the simplest and oldest thermochemical technology used very common in energy cogeneration systems. Due to high price of electricity, cogeneration facilities using bagasse can produce significant cost savings [23]. Cogeneration process using bagasse is also an alternative of saving water in the reservoirs of hydroelectric power plants [200]. Brazil is the top most country for electricity generation using sugarcane bagasse. As compared to 2010 data, Brazilian government projections exported 65% more electricity to the country's grid in 2019. The mills in Brazil generate about 12 kWh of electricity, 16 kWh of mechanical energy and 330 kWh of heat energy per ton of sugarcane bagasse [198]. India has been able to supply over 3000 MW of power to the country's national grid [202].

5.3. As animal feed

Due to shortage of roughage, poor livestock productivity and high cost of feed constituents, there is need of better feed and roughage for cattle which provides adequate nutrition. Sugarcane bagasse is used for ruminant feeding but it has high fibre and lignin content as compared to protein [203]. Due to high lignin, ruminal digestion of hemicellulose and cellulose is inhibited resulting in lower nutrition. Therefore,

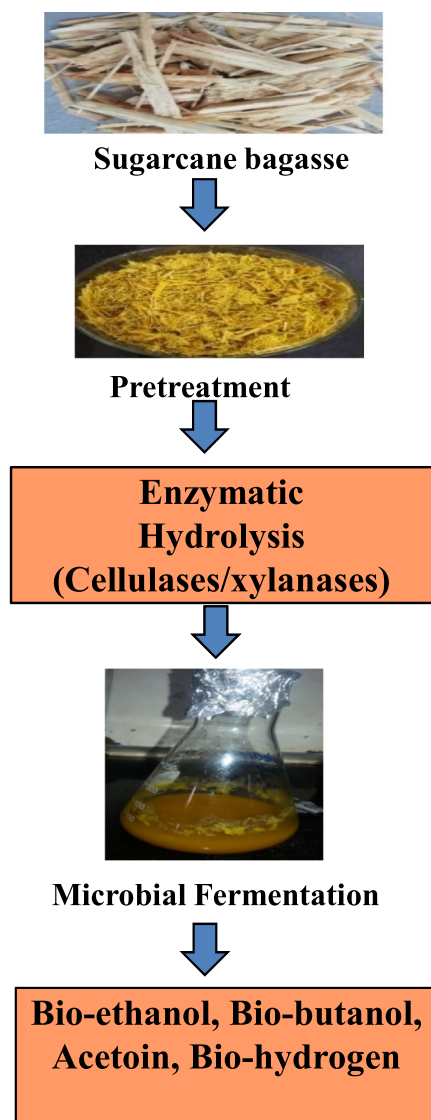


Fig. 4. Process for the production of biofuels using sugarcane bagasse.

nutritional value of sugarcane bagasse has been improved by biological, chemical and physical treatments to disrupt lignocellulosic complex [204]. Gunun et al. [205] revealed that sugarcane bagasse nutritive value increased after calcium hydroxide and urea treatment resulting in increased digestibility, rumen fermentation and feed intake. Lunsin et al. [174] reported that molasses and urea treatment increased the nutritional value for dairy cattles. Samadi et al. [173] reported high nutritional value of sugarcane bagasse after ammonification of sugarcane bagasse followed by fermentation with *Trichoderma harzianum*. Biological treatment of sugarcane bagasse with *Pleurotus florida* (oyster mushroom) increased the digestibility, relative palatability index and voluntary feed intake. Autoclaved and dried sugarcane bagasse promoted the digestion of rabbits [206]. Many studies reported enhancement of nutritional value of sugarcane bagasse after urea and various other treatments [207,208]. As a result of various treatments, sugarcane bagasse can be used as a good feed and roughage for cattles and farm animals. White-rot fungi, primarily basidiomycetes, are the major degraders of the lignified wood material and the end product formed is known as *Palo-podrido* which is used as animal feed [209]. Li et al. [210] reported that whole sugarcane bagasse was incubated with white rot fungi due to which lignin and hemicelluloses content was decayed and the fermented product was used as animal feed. All the

studies showed that the treated bagasse can be used as an alternative roughage and nutritional source for ruminant feeding in an economical manner [211].

5.4. Industrial enzymes

Sugarcane bagasse has tremendously been utilized for the production of various enzymes (cellulase, xylanase, lipase, amylase, phytase and others) using bacterial and fungal strain in solid state (SSF) and submerged fermentations (SmF) at industrial scale (Table 1). Following are the different enzymes produced by the microorganisms utilizing sugarcane bagasse as substrate (Fig. 5).

5.4.1. Xylanase

Xylanase is a hydrolytic enzyme which breakdown β -1,4 glycosidic bond of xylan backbone. It has gathered attention due to its application in various fields like paper and pulp, bakery products, biofuel production and food industries [212]. Sugarcane bagasse, cane molasses and other sugarcane products have been used for xylanase production. Alokika and Singh [163] reported high xylanase production (98.16 U/g substrate) by *Bacillus subtilis* subsp. *subtilis* JJBS250 using sugarcane bagasse under SSF after 48 h. Morais et al. [162] attained high xylanase production (0.284 U/ml) by *Aureobasidium pullulans* using sugarcane bagasse at 30 °C. High xylanase production (3400 U/ml) was observed in SSF by *Aspergillus niger* using sugarcane bagasse as a substrate [160]. Solid state fermentation of sugarcane bagasse by *Trichoderma viride*-IR05 resulted in 72.4 ± 1.42 U/g xylanase after 7 days [159]. Enhanced xylanase production (47.33 IU/ml) by *Bacillus subtilis* BS04 was reported using sugarcane bagasse in SmF [161].

5.4.2. Cellulase

Cellulase is a complex of enzymes (exoglucanase, endoglucanase and β -glucosidase) which hydrolyze cellulose into glucose units [213]. This enzyme is widely used in biofuel production from lignocellulosic substrates including bagasse. *Trichoderma koningii* produced maximum cellulase (8.2 U/g) using raw sugarcane bagasse as compared to pretreated bagasse in SSF [167]. Sugarcane bagasse was utilized for cellulase (3.7 FPU/ml) production in SmF [165]. Onofre et al. [166] observed high cellulase production (33.15 U/g) by *Aspergillus flavus* using sugarcane bagasse at 27 °C.

5.4.3. α -Amylase

Amylase converts starch into glucose and maltose. It has various applications in beverage, detergent, textile and starch industries. Production of amylase can be made economical by using low cost lignocellulosic biomass like sugarcane bagasse as substrate for the cultivation of microorganisms in SSF and SmF [214]. Rajagopalan and Krishnan [168] reported high α -amylase production (67.4 U/ml) from *Bacillus subtilis* using sugarcane bagasse.

5.4.4. Lipase

Lipase hydrolyzes fats into fatty acids and glycerol. Lipases are useful in many industrial applications like cosmetics, food, detergents, pharmaceutical and fuel industries. Thermostable lipase production has been reported under SSF for the valorization of bagasse [214]. Vaseghi et al. [170] investigated high lipase production (215 U/gds) using sugarcane bagasse in tray fermenter by *Rhizopus oryzae*. Higher lipase production (40 U/ml) by *Aspergillus fumigatus* was reported using pretreated sugarcane bagasse supplemented with mineral salts [171].

5.5. Sugars and oligosaccharides production

Large amount of sugarcane bagasse generated every year can be used for various purposes like oligosaccharides and sugars production. There are many studies regarding the use of bagasse in production of these valuable products. After enzymatic hydrolysis of sugarcane

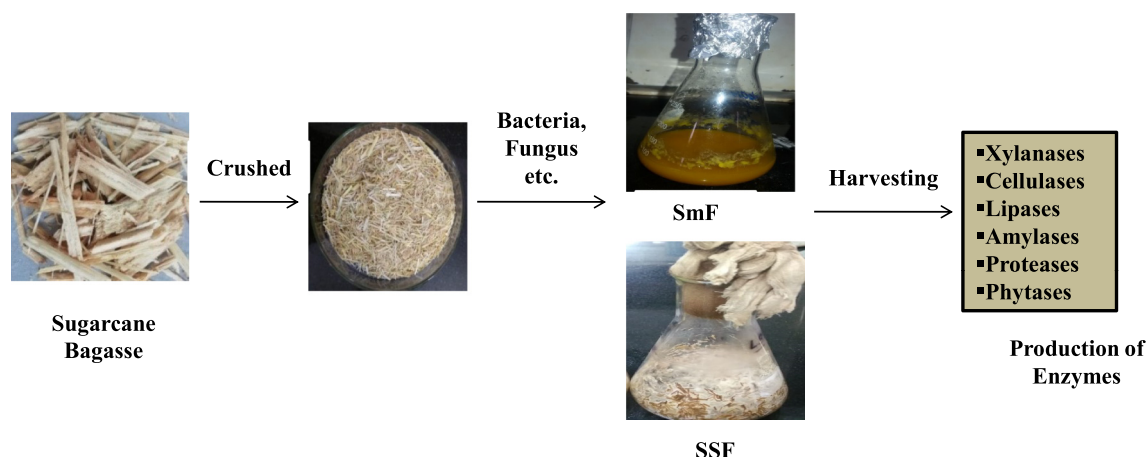


Fig. 5. Production of different enzymes using sugarcane bagasse.

bagasse, hemicelluloses and cellulose are converted into various useful fermentable sugars. These sugars are further used in various other applications like biofuel production. Xylooligosaccharides (XOS) are oligomers generated by hydrolyzing the xylan present in hemicellulosic fraction of bagasse. XOS are useful in various health benefits like boosting growth of beneficial gut bacteria, reduction in cholesterol levels, laxation, improving immunity and also used as prebiotics in food [176,178]. XOS can also be used as hydrogels and films for food packaging in its polymeric form. Presently, the market value of XOS is nearly \$22–50/kg [215].

Mandelli et al. [177] reported simultaneous production of antioxidant compounds and XOS from enzymatic hydrolysate of sugarcane bagasse. Combinatorial pretreatment of bagasse with acetic acid and hydrogen peroxide followed by saccharification using recombinant xylanase from *Bacillus subtilis* resulted in high yield of XOS [176]. Li et al. [216] revealed coproduction of fermentable sugars and XOS from the enzymatic hydrolysis of alkaline oxidation pretreated sugarcane bagasse. Zhou and Xu [178] found maximum XOS production from prehydrolyzed sugarcane bagasse. Ávila et al. [217] also reported XOS production from dilute sulfuric acid pretreated sugarcane bagasse. High sugar production (7.32 g/l) was attained from enzymatic saccharification of bagasse [182]. Untreated sugarcane bagasse liberated maximum reducing sugars (124.24 mg/g substrate) as compared to sodium carbonate pretreated biomass using the bacterial xylanase [163]. There is one report regarding sugar production after autohydrolysis and mechanical refining followed by enzymatic hydrolysis of sugarcane bagasse [218].

5.6. Used in gasification in gasifier system

Gasification is a most flexible thermo-chemical process used to produce clean energy by the conversion of solid biomass. In this process, any carbon containing biomass breaks down into its basic chemical constituents called as synthetic gas (syn-gas). The process is occurred by reacting the biomass at high temperatures, mainly above 1000 °C, in the presence of limited oxygen. Syngas consists of mainly CO, CO₂, H₂, CH₄ and N₂. The main aim of this process is utilization of waste biomass into useful energy [219]. Syngas can be used in electricity generation, stationary turbines or as a building block for a variety of fuels and chemicals. This syngas is differ from natural gas in terms of composition, flammability and heating value. During gasification, 1st step is the thermochemical conversion of sugarcane bagasse into cellulose, hemicellulose and lignin, which undergo depolymerization to gases and low molecular weight volatiles [220].

High cellulose content in biomass increases the rate of gasification with improved yields of gas and tar. CO and CO₂ are the main gaseous components produced from the decomposition of cellulose through

decarboxylation process. Hemicellulose reaction behaviour is similar to cellulose, whereas lignin decomposition requires higher temperature due to highly cross-linked polymer of phenoxy and methoxy substituted phenyl propane units. Most commonly used gasifier is downdraft because of its affordability, small scale production for the low power consumption in domestic factories [4]. Sugarcane bagasse is one of the most useful lignocellulosic biomass used in downdraft gasifier. The composition and moisture of bagasse also affects the gasification, therefore, preprocessing is carried out to overcome these limitations. Size reduction, pelletising, drying, torrefaction and briquetting are various methods used in the preprocessing of sugarcane bagasse [221]. Akay and Jordan [220] reported gasification of sugarcane bagasse in a downdraft gasifier. Small densified fuel particle size is necessary for syngas production because high ratio of cellulose/lignin, slower the rate of devolatilization.

5.7. As biosorbent for removal of hazardous material

Due to urbanization and industrialization, various non-biodegradable, hazardous materials like dyes, heavy metals and organic pollutants are responsible for environmental pollution and serious health problems. Various physicochemical processes like membrane filtration, coagulation, sedimentation and ion exchange have been used for the removal of these pollutants [222]. But now bagasse-derived adsorptive materials are being developed because it is produced luxuriantly every year, cheap, easily modifiable, highly sorptive and easily manipulative for treatment process. Bagasse is cost-effective and abundant agroresidue used as bioadsorbent [24,25]. Biosorption is the process of adsorption using agricultural biomass/residues. Sugarcane bagasse being a fibrous lignocellulosic material provides strong binding sites for pollutants because of the presence of various functional groups. Bagasse bioadsorbent is composed of macromolecules like hemicellulose, cellulose, lignin, fulvic, humic substances and proteins with many different functional groups (–COOH, –NH₂, –OH, –CONH₂, –SH₂, and –OCH₃) acting as adsorptive sites [223]. These bioadsorbent sites bind and attract pollutant ions either by donation of electron pairs, adsorption or by replacing them with hydrogen ions [29].

The adsorption efficiency of bagasse also depends upon various factors like temperature, pH, initial pollutant concentration, biosorbent dosage, sorbent particle size and contact time. Moubarik and Grimi [224] reported maximum (96%) removal of Cd(II) at pH 7.0 and 10 ppm Cd(II) concentration after 25 min of contact time. Raw sugarcane bagasse has been used for removing pollutants but physical and chemical treated bagasse is more suitable as biosorbent. Physical treatments include steaming, milling, grinding and autoclaving, while chemical treatment includes using chemicals, surfactants, chelating agents, polymerization, grafting and mercerization. EDTA-modified or activated

sugarcane bagasse has more adsorption capacity for Cr(VI) and Zn(II) than unmodified bagasse [225]. Dos Santos et al. [175] reported that adsorption capacity of citric acid modified sugarcane bagasse for Cu(II) increased to 31.53 mg/g as compared to raw bagasse (6.87 mg/g). Due to high amount of cellulose in sugarcane bagasse, it shows great capacity of metal biosorption. Biosorption of various metals like Ni, Cr, Pb, Cu, Cd, Hg has been reported using sugarcane bagasse [27]. do Carmo Ramos et al. [24] reported capacity of carboxylate functionalized sugarcane bagasse in removing nickel ions. Homagai et al. [226] found high absorption of copper(II) ions (184.92 mg/g) using xanthated sugarcane bagasse.

5.8. Sugarcane bagasse ash for stabilized earth blocks and tiles

Valorization of sugarcane bagasse refers to any industrial process activity aimed at recycling, reusing and composting of bagasse into sources of energy or useful products. Therefore, researchers are concentrating toward the use of bagasse for the manufacturing of blocks and bricks because of the shortage of traditionally used raw materials, labour and cost rising materials [227,228]. The ash is obtained from the bottom of the boilers and from the washing of the chimney gases [229]. The ash have both inorganic (silica) and organic (charcoal, bagasse debris) constituents, from which inorganic constituent can be used for the manufacturing of bricks [230] and ceramics [231]. Composition of bagasse ash includes alumina, iron oxide and silica. Some reports identified the presence of ferric oxide, calcite, cristobalite and quartz in the bagasse ash [232]. Schettino and Holanda [233] revealed the presence of potassium carbonate, mullite, hematite and calcium phosphate in addition to cristobalite and quartz. Pretreatment also helps in improving ash reactivity by milling or sieving process.

In recent years, bagasse ash has been utilized in manufacturing of stabilized earth blocks and sintered blocks. Bagasse ash addition to cement increased the performance of the stabilized blocks. Bagasse ash has increased the durability of stabilized blocks. Use of bagasse ash (25%) produced cement stabilized bricks with low production cost and high compressive strength [234]. Salim et al. [235] reported utilization of 10% bagasse ash in manufacture of sand loam soil compressed earth blocks and found 65% increase in the strength of blocks. Singh and Kumar [236] manufactured light weight bricks by the combination of sand, cement and bagasse ash. A combination of 20% cement and 15% bagasse ash produced highest strength of block with efflorescence and water absorption.

5.9. Commercial products from sugarcane bagasse

Various byproducts of sugarcane industry such as sugarcane bagasse and leaves are rich source of carbohydrates, having potential to generate value-added products of commercial importance. Bio-industrial utilization of bagasse offers environmental, sustainable and economical benefits in microbiology and biotechnology fields, thus providing enormous potential applications. In view of this, sugarcane bagasse has been successfully utilized for bioconversion into various valuable products i.e., industrial enzymes, ethanol, organic acid, xylitol and others. Along with biofuels, it has already been considered for various products of commercial/industrial importance which are given below:

5.9.1. Single cell protein

Single cell protein (SCP) usually refers to single cell dietary microbes derived from mushrooms, yeasts, micro-algae, and bacteria in pure or mixture form. These are known as protein rich ingredients/food or dietary supplements [237]. These are considered good alternative of agricultural protein as their production is characterized by less water consumption [238]. Finco et al. [239] elaborated the reported studies on production of single cell protein or oil from various waste by-products such as starch rich waste products (rice bran, wheat bran), structural polysaccharides rich biomass (wheat straw, bran, sugarcane

bagasse) and glycerol-enriched waste products. Samadi et al. [181] observed 13.41% SCP by utilizing NaOH pretreated sugarcane bagasse fermented by *S. cerevisiae* for 72 h. Similarly, Nigam [180] utilized sugarcane bagasse for effective production of single cell protein by *Candida langeronii* RLJY-019.

5.9.2. Organic acids

Bagasse has been used for the production of various organic acids like citric acid [240], lactic acid [241], gluconic [178] and succinic acid [242] [Table 2]. As compared to first generation feedstocks, bagasse derived organic acids can be cost effective [2].

5.9.2.1. Citric acid. Citric acid is considered as an important organic acid, enormously utilized in chemical, food and beverage industries [243]. But its excessive demand needs an alternative of utilizing low-cost lignocellulosic biomass to improve the efficacy of microbes [244]. Several reports described the use of various easily available lignocellulosic agroresidues as suitable substrates for the production of citric acid [245,246]. Therefore, sugarcane bagasse is a preferred substrate for production of citric acid related to its cost and production level in sugar mills [240]. Sugarcane bagasse has solid support characteristics of providing carbon or/and nutrients for the production of organic acids. Yadegary et al. [240] reported 87.32 g/kg and 97.81 g/kg citric acid using acid and sodium hydroxide pretreated sugarcane bagasse by *A. niger*. Bastos et al. [247] reported high citric acid production (17.5 g/kg of solid medium) using sugarcane bagasse and vinasse by *Aspergillus niger* after 3 days. Sugarcane bagasse with vinasse resulted in increased production of citric acid (14.5 g/kg of solid medium) by *Aspergillus niger* per day [248]. Campanhol et al. [249] utilized combination of sugarcane bagasse and vinasse and reported 734 mg/L citric acid using microbial consortium of *Trichoderma reesei* and *Aspergillus niger* after 48 h.

5.9.2.2. Lactic acid. Lactic acid, a valuable organic acid having extensive applications in pharmaceutical, chemical, food and textile industries. Huge demand of lactic acid has been grown substantially recently due to enormous potential for the production of biodegradable poly-lactic acid as well as eco-friendly agrobased materials i.e., acts as an alternative of oil based plastics such as polystyrene and polyethylene [250–252]. It is also used as intermediate for plastic and polymer industry [253]. Usually high sugar rich crops/sources (cane molasses, starch rich feed) have been preferred for microbial fermentation of lactic acid [254]. But the less availability of these crops seems to be the main challenge for current research in this field. Therefore, utilization of economical and luxuriantly available lignocellulosic biomass i.e., sugarcane bagasse can improve the economic feasibility for lactic acid production [241]. Wischral [241] utilized acid and alkaline pretreated sugarcane bagasse and reported 65.0 g/l lactic acid using *Lactobacillus pentosus* in batch fermentation. Similarly, Unrean [255] reported 72.7 g/l lactic acid by *Lactobacillus pentosus* using sugarcane bagasse in fed-batch fermentation. Edwin et al. [256] pretreated the bagasse with sulfuric acid and reported 70.4 g/l lactic acid in simultaneous saccharification and fermentation and 58.7 g/l lactic acid by a two-stage simultaneous saccharification and fermentation process. Peng et al. [153] reported 185.0 g/l lactic acid in sugarcane bagasse hydrolysate using *Bacillus* sp. P38 in fed-batch fermentation.

5.9.2.3. Poly-γ-glutamic acid. Poly-γ-glutamic acid (PGA), an anionic polyamide has gained research interest due to its extensive-properties of bitterness reliever and thickener agent, mineral adsorbent and cryoprotectant [257]. PGA is also preferred in food industry as texture enhancer, sedimenter, food supplement and hydrogel. Due to its chemical characteristics, it is also in huge demand in health care industry [258]. Furthermore, it exhibited high potential in agriculture (as a fertilizer synergist), medical (as biological adhesives), water treatment (as biopolymer flocculent) and so on. Anju et al. [259] studied the

utilization of various agrowastes such as sugarcane bagasse, rice straw and sorghum stover for the effective production of PGA. Feng et al. [260] reported 45.7 g/l PGA by immobilization of *Bacillus subtilis* NX-2 on chemically pretreated sugarcane bagasse.

5.9.3. Oil production

Microbial oil production using different monomeric carbon sources such as glucose, glycerol, and xylose is still a major issue of poor economics due to high cost of feedstock as well as low fermentation efficacy [261]. To improve the economic viability, low cost and easily available lignocellulosic substrate along with an ideal pretreatment can be considered for efficient production of microbial oil [262,263]. Various studies have also reported the utilization of pure or crude glycerol as well as combination of glycerol-glucose for the production of healthy fatty acid such as arachidonic acid, eicosapentaenoic acid and others [264]. Filamentous fungi and yeasts (*Rhodotorula*, *Yarrowia*, *Lipomyces* and *Rhodospiridium*) have been preferred as alternative oil source for the production of biodiesel and advanced fuels [265–268]. Among these, red yeast strains (*Rhodospiridium*, *Rhodotorula*) have attained researcher's interest due to their high potential to produce microbial oil along with carotenoids [269–271]. Cai, Moghaddam et al. [272] pretreated the sugarcane bagasse with acidified glycerol and reported 78.7% bio-oil using *Mortierella isabellina* NRRL 1757 after 6 days. Similarly, production of 3.9 g/L oil was reported by Bonturi et al. [264] using dilute acid pretreated bagasse. Hassanpour et al. [273] reported 7.7 g/L oil and 3.5 mg/L carotenoid by *Rhodospiridium toruloides* RP 15 using acidified glycerol pretreated sugarcane bagasse.

6. Sugarcane bagasse: an ideal and model substrate for biorefinery approach

Sugarcane bagasse a byproduct generated after processing of sugarcane acts as a potential source of carbohydrates at commercial scale [9]. Generally bagasse is considered as raw material for burning in sugarcane mill furnaces as its low calorific power proved this process as an effective and low efficiency approach [274]. Sugarcane bagasse contains low amount of ash, hence offers various advantages as compared to other residues (rice straw, wheat straw) in bioconversion process at large scale. Secondly, sugarcane bagasse acts as an abundant solar energy source due to luxuriant production and high regeneration capacity as compared to other lignocellulosic residues such as grasses, wheat straw, rice straw and others [30]. Bagasse fibres play an important role in polymers reinforcement as their burning result in generation of ash rich in inorganic components that has been used in the synthesis of high strength bricks and tiles [7]. Presence of silica in bagasse ash is an effective raw material for improving the strength and stability of building materials. Bagasse requires low cost input as byproduct is used as an effective fertilizer for improving soil fertility [3].

Sugarcane bagasse acts as a potent feedstock for biofuel industry as it is luxuriantly available and sustainable biomass. Further, the presence of relatively high carbohydrates and low lignin in raw bagasse, makes it an efficient substrate for enhanced saccharification followed by generation of different value-added products [163,275]. Both cellulose and hemicellulose of bagasse are enzymatically hydrolyzed into monomeric sugars (glucose and xylose), which are subsequently converts into biofuels and other value-added products by microbial fermentations [276]. As sugarcane bagasse is usually preferred for energy cogeneration, however, it could also be considered in production of various chemicals and fuels [277]. Ash generated due to the burning of bagasse has been effectively utilized for the synthesis of light weight and high strength bricks and tiles as building materials. Therefore, due to numerous advantages, bagasse is not only considered as a valuable byproduct of sugar industries but also acts as co-product having high added-benefits.

7. Conclusions and future prospects

A large amount of sugarcane bagasse is generated as byproduct in sugarcane industries all over the world as a result of sugarcane processing. A large part of bagasse is burnt in open by the sugar mills that is responsible for causing environmental pollution. Use of sustainable bagasse-management practices avoid the open burning and combat environmental pollution. There are various hazardous materials released into the environment by urbanization and industrialization, which cause threats to flora and fauna due to increase in chemical based pollution. The environmental and energy crises are forcing the world to re-evaluate the efficient and economic alternatives for utilizing the lignocellulosic biomass including bagasse. The application of lignocellulosic waste materials including sugarcane bagasse is a cleaner approach for clean and green energy production as well as for reduction and proper utilization of these wastes. Sustainable and efficient utilization of a lignocellulosic biomass depends on degree of polymerization, cellulose and hemicellulose content and moisture content as well as on the availability of biomass, thickness, lignin content and others. Sugarcane bagasse has all the requisite properties as a sustainable alternative biomass that has been utilized in the production of biofuels, electricity generation, enzymes, sugars, organic acids, animal feed, biosorbent and other value-added products. Sugarcane bagasse has high amount of cellulose and hemicellulose along with lignin and silica. Presence of lignin and silica make the structure more rigid and recalcitrant. The recalcitrancy of biomass demands an ideal pretreatment to enhance the productivity that could minimize the microorganisms and enzymes cost to obtain high titres of respective product. Therefore, pretreatment of bagasse is the necessary requirement to overcome the barriers of enzymatic hydrolysis. Different physical and chemical methods used for the pretreatment of sugarcane bagasse are also responsible for creating further pollution problems in the environment. Therefore, development of an environment-benign process using microorganisms (specifically basidiomycetous fungi) and their enzymes could be the sustainable and cleaner alternatives for the biorefinery approach.

Sugarcane bagasse has been utilized as nutrient medium for the cultivation of various microorganisms for the production of different value-added products including single cell protein, enzymes, organic acids, prebiotics and reducing sugars. Resulting sugars have been used in microbial fermentation for the production of 2nd generation biofuels like bioethanol, biobutanol, biohydrogen and others. Residual unhydrolyzed biomass can be utilized as green manure for improving the soil fertility and enhancing the crop yield. Sugarcane bagasse has been used directly for generating electricity, syngas and cattle feed. Bagasse is also used as biosorption agent because of unique binding sites which help in adsorption of various hazardous materials. Resultant ash generated due to burning of bagasse has been used in the synthesis of high strength bricks and tiles. In the current scenario, future trends are being directed toward the economic use of sugarcane bagasse. Future research should be focused in improving the efficiency of bagasse utilization and cogeneration for sustainable biofuels and other products. Biological pretreatment of sugarcane bagasse should be carried out in order to make the pretreatment process economical and eco-friendly. Sugarcane bagasse may be explored for newer products like acetoin, prebiotics and bioplastics. Lignocellulosic biomass and green chemistry should be the main thrust areas of future research to mitigate the energy crisis and environmental pollution. Hence there is need of global awareness regarding clean technology as well as research efforts to propose feasible, economical and eco-friendly clean alternatives for the sustainable utilization of sugarcane bagasse.

Declaration of competing interest

The authors declare no conflict of interest.

Acknowledgements

Alokika is grateful to the Council of Scientific and Industrial Research (CSIR), New Delhi, for financial assistance (No. 09/382(0179)/2016-EMR1) during this research work. BS and Anu are thankful to the Haryana State Council for Science and Technology, Panchkula for providing financial support (HSCST/R&D/2017/62; No. 1743, dated April 12, 2017) for research during writing of this review.

References

- [1] T. Silalertruksa, S.H. Gheewala, Competitive use of sugarcane for food, fuel, and biochemical through the environmental and economic factors, *Int. J. Life Cycle Assess.* (2019) 1–13.
- [2] M. Meghana, Y. Shastri, Sustainable valorization of sugar industry waste: status, opportunities, and challenges, *Bioresour. Technol.* 303 (2019), 122929.
- [3] M.L. Dotaniya, S.C. Datta, D.R. Biswas, C.K. Dotaniya, B.L. Meena, S. Rajendran, K.L. Regar, M. Lata, Use of sugarcane industrial by-products for improving sugarcane productivity and soil health, *Int. J. Recycl. Org. Waste Agric.* 5 (2016) 185–194.
- [4] A. Anukam, S. Mamphweli, P. Reddy, E. Meyer, O. Okoh, Pre-processing of sugarcane bagasse for gasification in a downdraft biomass gasifier system: a comprehensive review, *Renew. Sust. Energ. Rev.* 66 (2016) 775–801.
- [5] S. Yao, S. Nie, Y. Yuan, S. Wang, C. Qin, Efficient extraction of bagasse hemicelluloses and characterization of solid remainder, *Bioresour. Technol.* 185 (2015) 21–27.
- [6] S. Al Arni, Extraction and isolation methods for lignin separation from sugarcane bagasse: a review, *Ind. Crop. Prod.* 115 (2018) 330–339.
- [7] T.C. Mokheba, M.J. Mochane, T.E. Motaung, L.Z. Linganisio, O.M. Thekisoe, S.P. Songca, Sugarcane bagasse and cellulose polymer composites, *Sugarcane-Technology and Research*, IntechOpen 2017, pp. 225–240.
- [8] G.R. Timilsina, D. Zilberman, *Impacts of Biofuels on the Economy, Environment, and Poverty*, Springer-Verlag New York, 2016.
- [9] S.C. Rabelo, A.C. da Costa, C.E.V. Rossel, Industrial waste recovery, *Sugarcane*, Academic Press 2015, pp. 365–381.
- [10] R.L.L. Ázar, T. Morgan, M.H. Barbosa, V.M. Guimarães, E. Ximenes, M. Ladisch, Impact of protein blocking on enzymatic saccharification of bagasse from sugarcane clones, *Biotechnol. Bioeng.* 116 (2019) 1584–1593.
- [11] D. Orrego, A.D. Zapata-Zapata, D. Kim, Optimization and scale-up of coffee mucilage fermentation for ethanol production, *Energies* 11 (2018) 786.
- [12] M.O. Dias, M.P. da Cunha, R. Maciel Filho, A. Bonomi, C.D. Jesus, C.E. Rossell, Simulation of integrated first and second generation bioethanol production from sugarcane: comparison between different biomass pretreatment methods, *J. Ind. Microbiol. Biotechnol.* 38 (2011) 955–966.
- [13] S. Macrelli, J. Mogensen, G. Zacchi, Techno-economic evaluation of 2nd generation bioethanol production from sugar cane bagasse and leaves integrated with the sugar-based ethanol process, *Biotechnol. Biofuels* 5 (2012) 22.
- [14] T.L. Bezerra, A.J. Ragauskas, A review of sugarcane bagasse for second-generation bioethanol and biopower production, *Biofuels Bioprod. Biorefin.* 10 (2016) 634–647.
- [15] M. Gavrilescu, Biomass potential for sustainable environment, biorefinery products and energy, *Sustainable Energy in the Built Environment-Steps Towards nZEB*, Springer, Cham 2014, pp. 169–194.
- [16] M. Pande, A.N. Bhaskarwar, Biomass conversion to energy, *Biomass Conversion*, Springer 2012, pp. 1–90.
- [17] C. Florencio, A.C. Badino, C.S. Farinas, Addition of soybean protein improves saccharification and ethanol production from hydrothermally pretreated sugarcane bagasse, *Bioenergy Res* 12 (2019) 81–93.
- [18] D. Kim, Physico-chemical conversion of lignocellulose: inhibitor effects and detoxification strategies: a mini review, *Molecules* 23 (2018) 309.
- [19] B.S. Donohoe, T.B. Vinzant, R.T. Elander, V.R. Pallapolu, Y.Y. Lee, R.J. Garlock, V. Balan, B.E. Dale, Y. Kim, N.S. Mosir, M.R. Ladisch, Surface and ultrastructural characterization of raw and pretreated switchgrass, *Bioresour. Technol.* 102 (2011) 11097–11104.
- [20] D. Kim, S. Ku, *Bacillus* cellulase molecular cloning, expression, and surface display on the outer membrane of *Escherichia coli*, *Molecules* 23 (2018) 503.
- [21] G.P. Maitan-Alfenas, E.M. Visser, V.M. Guimarães, Enzymatic hydrolysis of lignocellulosic biomass: converting food waste in valuable products, *Curr. Opin. Food Sci.* 1 (2015) 44–49.
- [22] A.V. Rodhe, L. Sateesh, J. Sridevi, B. Venkateswarlu, L.V. Rao, Enzymatic hydrolysis of sorghum straw using native cellulase produced by *T. reesei* NCIM 992 under solid state fermentation using rice straw, *Biotech* 1 (2011) 207–215.
- [23] M. Arshad, S. Ahmed, Cogeneration through bagasse: a renewable strategy to meet the future energy needs, *Renew. Sust. Energ. Rev.* 54 (2016) 732–737.
- [24] S.N. do Carmo Ramos, A.L.P. Xavier, F.S. Teodoro, M.M.C. Elias, F.J. Gonçalves, L.F. Gil, R.P.D. Freitas, L.V.A. Gurgel, Modeling mono- and multi-component adsorption of cobalt (II), copper (II), and nickel (II) metal ions from aqueous solution onto a new carboxylated sugarcane bagasse. Part I: batch adsorption study, *Ind. Crop. Prod.* 74 (2015) 357–371.
- [25] S.N. do Carmo Ramos, A.L.P. Xavier, F.S. Teodoro, L.F. Gil, L.V.A. Gurgel, Removal of cobalt (II), copper (II), and nickel (II) ions from aqueous solutions using phthalate-functionalized sugarcane bagasse: mono- and multicomponent adsorption in batch mode, *Ind. Crop. Prod.* 79 (2016) 116–130.
- [26] P. Singh, S.N. Singh, A.K. Tiwari, S.K. Pathak, A.K. Singh, S. Srivastava, N. Mohan, Integration of sugarcane production technologies for enhanced cane and sugar productivity targeting to increase farmers' income: strategies and prospects, *3 Biotech* 9 (2019), 48.
- [27] R.Z. Khoo, W.S. Chow, H. Ismail, Sugarcane bagasse fiber and its cellulose nanocrystals for polymer reinforcement and heavy metal adsorbent: a review, *Cellulose* 25 (2018) 4303–4330.
- [28] A. Balaji, B. Karthikeyan, C.S. Raj, Bagasse fiber—the future biocomposite material: a review, *Int. J. ChemTech Res.* 7 (2014) 223–233.
- [29] G.J. de Moraes Rocha, V.M. Nascimento, A.R. Gonçalves, V.F.N. Silva, C. Martín, Influence of mixed sugarcane bagasse samples evaluated by elemental and physical-chemical composition, *Ind. Crop. Prod.* 64 (2015) 52–58.
- [30] A. Pandey, C.R. Soccol, P. Nigam, V.T. Soccol, Biotechnological potential of agro-industrial residues. I: sugarcane bagasse, *Bioresour. Technol.* 74 (2000) 69–80.
- [31] W. Xiong, Bagasse composites: a review of material preparation, attributes, and affecting factors, *J. Thermoplast. Compos. Mater.* 31 (2018) 1112–1146.
- [32] A.H. Hemmasi, I. Ghasemi, B. Bazyar, A. Samariha, Studying the effect of size of bagasse and nanoclay particles on mechanical properties and morphology of bagasse flour/recycled polyethylene composites, *BioResources* 8 (2013) 3791–3801.
- [33] Y. Cao, K. Goda, S. Shibata, Development and mechanical properties of bagasse fiber reinforced composites, *Adv. Compos. Mater.* 16 (2007) 283–298.
- [34] M. Chaharmahali, Y. Hamzeh, G. Ebrahimi, A. Ashori, I. Ghasemi, Effects of nanographene on the physico-mechanical properties of bagasse/polypropylene composites, *Polym. Bull.* 71 (2014) 337–349.
- [35] A. Samariha, A. Bastani, M. Nemati, M. Kiaei, H. Nosrati, M. Farsi, Investigation of the mechanical properties of bagasse flour/polypropylene composites, *Mech. Compos. Mater.* 49 (2013) 447–454.
- [36] G. Saini, A.K. Narula, V. Choudhary, R. Bhardwaj, Effect of particle size and alkali treatment of sugarcane bagasse on thermal, mechanical, and morphological properties of PVC-bagasse composites, *J. Reinf. Plast. Compos.* 29 (2010) 731–740.
- [37] A.A. El-Fattah, A.G.M. El-Demerdash, W.A. AlimSadik, A. Bedir, The effect of sugarcane bagasse fiber on the properties of recycled high density polyethylene, *J. Compos. Mater.* 49 (2015) 3251–3262.
- [38] R. Wirawan, S.M. Sapuan, Y. Robiah, A. Khalina, Elastic and viscoelastic properties of sugarcane bagasse-filled poly (vinyl chloride) composites, *J. Therm. Anal. Calorim.* 103 (2011) 1047–1053.
- [39] S.M. Zabihzadeh, S.K. Hosseini Hashemi, H. Mehregan Nikoo, S.M.J. Sepidehdam, Influence of fungal decay on physico-mechanical properties of a commercial extruded bagasse/PP composite, *J. Reinf. Plast. Compos.* 29 (2010) 1750–1756.
- [40] R.C. Kuhad, S. Kuhar, M. Kapoor, K.K. Sharma, A. Singh, Lignocellulosic microorganisms, their enzymes and possible biotechnologies based on lignocellulosic microorganisms and their enzymes, *Lignocellulose Biotechnology: Future Prospects*, IK International Pvt Ltd 2007, pp. 3–22.
- [41] A. Wali, M. Gupta, S. Gupta, V. Sharma, R.K. Salgotra, M. Sharma, Lignin degradation and nutrient cycling by white rot fungi under the influence of pesticides, *3 Biotech* 10 (2020) 266.
- [42] E. Nadali, A. Karimi, M. Tajvidi, R. Naghdi, Natural durability of a bagasse fiber/polypropylene composite exposed to rainbow fungus (*Coriolus versicolor*), *Reinf. Plast. Compos.* 29 (2010) 1028–1037.
- [43] F. Hemmati, H. Garmabi, A study on fire retardancy and durability performance of bagasse fiber/polypropylene composite for outdoor applications, *J. Thermoplast. Compos. Mater.* 26 (2013) 1041–1056.
- [44] M.G. Lomeli-Ramírez, H.G. Ochoa-Ruiz, F.J. Fuentes-Talavera, S. García-Enríquez, M.A. Cerpa-Gallegos, J.A. Silva-Guzmán, Evaluation of accelerated decay of wood plastic composites by *Xylophagus* fungi, *Int. Biodeterior. Biodegradation* 63 (2009) 1030–1035.
- [45] A. Samariha, A.H. Hemmasi, I. Ghasemi, B. Bazyar, M. Nemati, Effect of nanoclay contents on properties, of bagasse flour/reprocessed high density polyethylene/nanoclay composites, *Maderas. Ciencia y tecnología* 17 (2015) 637–646.
- [46] P. Harmsen, W. Huijgen, L. Lopez, R. Bakker, Literature Review of Physical and Chemical Pretreatment Processes for Lignocellulosic Biomass Energy Research Centre of the Netherlands, Wageningen University, 2010 Report.
- [47] N. Akhtar, D. Goyal, A. Goyal, Biodegradation of cellulose and agricultural waste material, in: Ram Chandra (Ed.), Chapter in Advances Unbiodegradation and Bioremediation of Industrial Waste, CRC Press, 2015, (ISBN-13: 978-1-4987-0055-9).
- [48] S. Al Arni, A.F. Drake, M. Del Borghi, A. Converti, Study of aromatic compounds derived from sugarcane bagasse. Part I: effect of pH, *Chem Eng Technol. Ind. Chem. Plant Equip. Proc. Eng. Biotech* 33 (2010) 895–901.
- [49] S. Al Arni, A.F. Drake, M. Del Borghi, A. Converti, Study of aromatic compounds derived from sugarcane bagasse: II. Effect of concentration, *Chem. Eng Technol. Ind. Chem. Plant Equip. Proc. Eng.-Biotech.* 33 (2010) 523–531.
- [50] A. Guarino Lino, J.L. Colodette, J.C.D. Rio Andrade, A. Gutiérrez Suárez, J. Rencoret, C. Lima, A.T. Martínez, A structural characterization of the lignins from sugarcane (*Saccharum* spp. L.) bagasse and straw, 7th International Colloquium on Eucalyptus Pulp, May 26–29, 2015, Vitoria, Espírito Santo, Brazil.
- [51] A. Brandt, J. Gräsvik, J.P. Hallett, T. Welton, Deconstruction of lignocellulosic biomass with ionic liquids, *Green Chem.* 15 (2013) 550–583.
- [52] W. Doherty, P. Halley, L. Edye, D. Rogers, F. Cardona, Y. Park, T. Woo, Studies on polymers and composites from lignin and fiber derived from sugar cane, *Polym. Adv. Technol.* 18 (2007) 673–678.
- [53] H. Boussarsar, B. Rogé, M. Mathlouthi, Optimization of sugarcane bagasse conversion by hydrothermal treatment for the recovery of xylose, *Bioresour. Technol.* 100 (2009) 6537–6542.
- [54] P. Laopaiboon, A. Thani, V. Leelavatcharamas, L. Laopaiboon, Acid hydrolysis of sugarcane bagasse for lactic acid production, *Bioresour. Technol.* 101 (2010) 1036–1043.

- [55] P.S. Ortiz, S. de Oliveira Jr., Exergy analysis of pretreatment processes of bioethanol production based on sugarcane bagasse, *Energy* 76 (2014) 130–138.
- [56] M.E. Vallejos, M.D. Zambon, M.C. Area, A.A. da Silva Curvelo, Low liquid-solid ratio fractionation of sugarcane bagasse by hot water autohydrolysis and organosolv delignification, *Ind. Crop. Prod.* 65 (2015) 349–353.
- [57] L.J. Jönsson, B. Alriksson, N.O. Nilvebrant, Bioconversion of lignocellulose: inhibitors and detoxification, *Biotechnol. Biofuels* 6 (2013) 16.
- [58] A. Barakat, C. Mayer-Laigle, A. Solhy, R.A. Arancón, H. De Vries, R. Luque, Mechanical pretreatments of lignocellulosic biomass: towards facile and environmentally sound technologies for biofuels production, *RSC Adv.* 4 (2014) 48109–48127.
- [59] A. Barakat, H. de Vries, X. Rouau, Dry fractionation process as an important step in current and future lignocellulose biorefineries: a review, *Bioresour. Technol.* 134 (2013) 362–373.
- [60] M.H.L. Silveira, A.R.C. Morais, A.M. da Costa Lopes, D.N. Oleksyszyn, R. Bogel-Lukasik, J. Andreus, L. Pereira Ramos, Current pretreatment technologies for the development of cellulosic ethanol and biorefineries, *ChemSusChem* 8 (2015) 3366–3390.
- [61] T. Gan, Y. Zhang, Y. Su, H. Hu, A. Huang, Z. Huang, D. Chen, M. Yang, J. Wu, Esterification of bagasse cellulose with metal salts as efficient catalyst in mechanical activation-assisted solid phase reaction system, *Cellulose* 24 (2017) 5371–5387.
- [62] X. Zhao, Y. Zhang, H. Hu, Z. Huang, M. Yang, D. Chen, K. Huang, A. Huang, X. Qin, Z. Feng, Effect of mechanical activation on structure changes and reactivity in further chemical modification of lignin, *Int. J. Biol. Macromol.* 91 (2016) 1081–1089.
- [63] Y. Zheng, J. Zhao, F. Xu, Y. Li, Pretreatment of lignocellulosic biomass for enhanced biogas production, *Prog. Energy Combust. Sci.* 42 (2014) 35–53.
- [64] M. Yaldagard, S.A. Mortazavi, F. Tabatabaie, The effect of ultrasound in combination with thermal treatment on the germinated barley's alpha-amylase activity, *Korean J. Chem. Eng.* 25 (2008) 517–523.
- [65] M.R. Esfahani, M. Azin, Pretreatment of sugarcane bagasse by ultrasound energy and dilute acid, *Asia Pac. J. Chem. Eng.* 7 (2012) 274–278.
- [66] R. Velmurugan, K. Muthukumar, Ultrasound-assisted alkaline pretreatment of sugarcane bagasse for fermentable sugar production: optimization through response surface methodology, *Bioresour. Technol.* 112 (2012) 293–299.
- [67] A.K. Kumar, S. Sharma, Recent updates on different methods of pretreatment of lignocellulosic feedstocks: a review, *Bioresour. Bioprocess* 4 (2017), 7.
- [68] T.C. Sarker, S.M.G.G. Azam, G. Bonanomi, Recent advances in sugarcane industry solid by-products valorization, *Waste Biomass Valorization* 8 (2017) 241–266.
- [69] P. Portero-Barahona, E.J. Carvajal-Barriga, J. Martín-Gil, P. Martín-Ramos, Sugarcane bagasse hydrolysis enhancement by microwave-assisted sulfonate pretreatment, *Energies* 12 (2019) 1703.
- [70] M.M. de Souza Moretti, D.A. Bocchini-Martins, C.D.C.C. Nunes, M.A. Villena, O.M. Perrone, R. da Silva, M. Boscolo, E. Gomes, Pretreatment of sugarcane bagasse with microwaves irradiation and its effects on the structure and on enzymatic hydrolysis, *Appl. Energy* 122 (2014) 189–195.
- [71] S. Kumar, P. Dheeran, S.P. Singh, I.M. Mishra, D.K. Adhikari, Kinetic studies of two-stage sulphuric acid hydrolysis of sugarcane bagasse, *Renew. Energy* 83 (2015) 850–858.
- [72] J. Shi, M.A. Ebrik, C.E. Wyman, Sugar yields from dilute sulfuric acid and sulfur dioxide pretreatments and subsequent enzymatic hydrolysis of switchgrass, *Bioresour. Technol.* 102 (2011) 8930–8938.
- [73] C.E. Wyman, B.E. Dale, R.T. Elander, M. Holtzapple, M.R. Ladisch, Y.Y. Lee, Comparative sugar recovery data from laboratory scale application of leading pretreatment technologies to corn stover, *Bioresour. Technol.* 96 (2005) 2026–2032.
- [74] L. Canilha, V.T. Santos, G.J. Rocha, J.B.A. de Silva, M. Giuliatti, S.S. Silva, M.G. Felipe, A.M. Milagres, W. Carvalho, A study on the pretreatment of a sugarcane bagasse sample with dilute sulfuric acid, *J. Ind. Microbiol. Biotechnol.* 38 (2011) 1467–1475.
- [75] X. Zhao, Y. Zhou, D. Liu, Kinetic model for glycan hydrolysis and formation of monosaccharides during dilute acid hydrolysis of sugarcane bagasse, *Bioresour. Technol.* 105 (2012) 160–168.
- [76] L. Qin, Z.H. Liu, B.Z. Li, B.E. Dale, Y.J. Yuan, Mass balance and transformation of corn stover by pretreatment with different dilute organic acids, *Bioresour. Technol.* 112 (2012) 319–326.
- [77] H. Zhang, Y. Xu, S. Yu, Co-production of functional xylooligosaccharides and fermentable sugars from corn cob with effective acetic acid prehydrolysis, *Bioresour. Technol.* 234 (2017) 343–349.
- [78] J. Bian, F. Peng, X.P. Peng, X. Xiao, P. Peng, F. Xu, R.C. Sun, Effect of [Emim] Ac pretreatment on the structure and enzymatic hydrolysis of sugarcane bagasse cellulose, *Carbohydr. Polym.* 100 (2014) 211–217.
- [79] A.M.J. Kootstra, H.H. Beentink, E.L. Scott, J.P. Sanders, Comparison of dilute mineral and organic acid pretreatment for enzymatic hydrolysis of wheat straw, *Biochem. Eng. J.* 46 (2009) 126–131.
- [80] X. Zhou, J. Zhao, X. Zhang, Y. Xu, An eco-friendly biorefinery strategy for xylooligosaccharides production from sugarcane bagasse using cellulosic derived gluconic acid as efficient catalyst, *Bioresour. Technol.* 289 (2019), 121755.
- [81] R. Timung, N. Naik Deshivath, V.V. Goud, V.V. Dasu, Effect of subsequent dilute acid and enzymatic hydrolysis on reducing sugar production from sugarcane bagasse and spent *Citronella* biomass, *J. Energy* 2016 (2016).
- [82] Q. Yu, X. Zhuang, S. Lv, M. He, Y. Zhang, Z. Yuan, W. Qi, Q. Wang, W. Wang, X. Tan, Liquid hot water pretreatment of sugarcane bagasse and its comparison with chemical pretreatment methods for the sugar recovery and structural changes, *Bioresour. Technol.* 129 (2013) 592–598.
- [83] S.C. Rabelo, H. Carrere, R. Maciel Filho, A.C. Costa, Production of bioethanol, methane and heat from sugarcane bagasse in a biorefinery concept, *Bioresour. Technol.* 102 (2011) 7887–7895.
- [84] L. Wu, Y. Li, M. Arakane, M. Ike, M. Wada, Y. Terajima, S. Ishikawa, K. Tokuyasu, Efficient conversion of sugarcane stalks into ethanol employing low temperature alkali pretreatment method, *Bioresour. Technol.* 102 (2011) 11183–11188.
- [85] L.L. Fuentes, S.C. Rabelo, R. Maciel Filho, A.C. Costa, Kinetics of lime pretreatment of sugarcane bagasse to enhance enzymatic hydrolysis, *Appl. Biochem. Biotechnol.* 163 (2011) 612–625.
- [86] G.A. Aita, D.A. Salvi, M.S. Walker, Enzyme hydrolysis and ethanol fermentation of dilute ammonia pretreated energy cane, *Bioresour. Technol.* 102 (2011) 4444–4448.
- [87] N. Yu, L. Tan, Z.Y. Sun, H. Nishimura, S. Takei, Y.Q. Tang, K. Kida, Bioethanol from sugarcane bagasse: focused on optimum of lignin content and reduction of enzyme addition, *Waste Manag.* 76 (2018) 404–413.
- [88] H. Zhang, S. Huang, W. Wei, J. Zhang, J. Xie, Investigation of alkaline hydrogen peroxide pretreatment and Tween 80 to enhance enzymatic hydrolysis of sugarcane bagasse, *Biotechnol. Biofuels* 12 (2019), 107.
- [89] G. Siqueira, A. Várnai, A. Ferraz, A.M. Milagres, Enhancement of cellulose hydrolysis in sugarcane bagasse by the selective removal of lignin with sodium chlorite, *Appl. Energy* 102 (2013) 399–402.
- [90] L. Moghaddam, Z. Zhang, R.M. Wellard, J.P. Bartley, I.M. O'Hara, W.O. Doherty, Characterisation of lignins isolated from sugarcane bagasse pretreated with acidified ethylene glycol and ionic liquids, *Biomass Bioenergy* 70 (2014) 498–512.
- [91] S. Bahrani, S. Raissi, M. Sarshar, Experimental investigation of ionic liquid pretreatment of sugarcane bagasse with 1, 3-dimethylimidazolium dimethyl phosphate, *Bioresour. Technol.* 185 (2015) 411–415.
- [92] M.L.W. Yoon, G.C. Ngoh, A.S. May Chua, M.A. Hashim, Comparison of ionic liquid, acid and alkali pretreatments for sugarcane bagasse enzymatic saccharification, *J. Chem. Technol. Biotechnol.* 86 (2011) 1342–1348.
- [93] D.M. de Carvalho, O. Sevastyanova, L.S. Penna, B.P. da Silva, M.E. Lindström, J.L. Colodette, Assessment of chemical transformations in *eucalyptus*, sugarcane bagasse and straw during hydrothermal, dilute acid, and alkaline pretreatments, *Ind. Crop. Prod.* 73 (2015) 118–126.
- [94] M. Galbe, G. Zacchi, Pretreatment: the key to efficient utilization of lignocellulosic materials, *Biomass Bioenergy* 46 (2012) 70–78.
- [95] L.J. Jönsson, C. Martín, Pretreatment of lignocellulose: formation of inhibitory by-products and strategies for minimizing their effects, *Bioresour. Technol.* 199 (2016) 103–112.
- [96] P. Moodley, E.G. Kana, Development of a steam or microwave-assisted sequential salt-alkali pretreatment for lignocellulosic waste: effect on delignification and enzymatic hydrolysis, *Energy Convers. Manag.* 148 (2017) 801–808.
- [97] M.C. de Albuquerque Wanderley, C. Martín, G.J. de Moraes Rocha, E.R. Gouveia, Increase in ethanol production from sugarcane bagasse based on combined pretreatments and fed-batch enzymatic hydrolysis, *Bioresour. Technol.* 128 (2013) 448–453.
- [98] R.S. Aguiar, M.H.L. Silveira, A.P. Pitarello, M.L. Corazza, L.P. Ramos, Kinetics of enzyme-catalyzed hydrolysis of steam-exploded sugarcane bagasse, *Bioresour. Technol.* 147 (2013) 416–423.
- [99] M.H.L. Silveira, A.K. Chandel, B.A. Vanelli, K.S. Sacilotto, E.B. Cardoso, Production of hemicellulosic sugars from sugarcane bagasse via steam explosion employing industrially feasible conditions: pilot scale study, *Bioresour. Technol. Rep.* 3 (2018) 138–146.
- [100] A.P. Pitarello, C.S.D. Fonseca, L.M. Chiarello, F.M. Gírio, L.P. Ramos, Ethanol production from sugarcane bagasse using phosphoric acid-catalyzed steam explosion, *J. Braz. Chem. Soc.* 27 (2016) 1889–1898.
- [101] S.C. Rabelo, R.R. Andrade, R. Maciel Filho, A.C. Costa, Alkaline hydrogen peroxide pretreatment, enzymatic hydrolysis and fermentation of sugarcane bagasse to ethanol, *Fuel* 136 (2014) 349–357.
- [102] K. Zhang, Z. Pei, D. Wang, Organic solvent pretreatment of lignocellulosic biomass for biofuels and biochemicals: a review, *Bioresour. Technol.* 199 (2016) 21–33.
- [103] B.W. Koo, H.Y. Kim, N. Park, S.M. Lee, H. Yeo, I.G. Choi, Organosolv pretreatment of *Liriodendron tulipifera* and simultaneous saccharification and fermentation for bioethanol production, *Biomass Bioenergy* 35 (2011) 1833–1840.
- [104] M.N. Borand, F. Karaosmanoglu, Effects of organosolv pretreatment conditions for lignocellulosic biomass in biorefinery applications: a review, *J. Renew. Sustain. Ener.* 10 (2018), 033104.
- [105] M.F. Li, S.N. Sun, F. Xu, R.C. Sun, Organosolv fractionation of lignocelluloses for fuels, chemicals and materials: a biorefinery processing perspective, *Biomass Conversion, Springer, Berlin, Heidelberg* 2012, pp. 341–379.
- [106] M.E. Santo, C.A. Rezende, O.D. Bernardinelli, N. Pereira Jr., A.A. Curvelo, E.R. Deazevedo, F.E. Guimarães, I. Polikarpov, Structural and compositional changes in sugarcane bagasse subjected to hydrothermal and organosolv pretreatments and their impacts on enzymatic hydrolysis, *Ind. Crop. Prod.* 113 (2018) 64–74.
- [107] H. Zhang, S. Wu, Efficient sugar release by acetic acid ethanol-based organosolv pretreatment and enzymatic saccharification, *J. Agric. Food Chem.* 62 (2014) 11681–11687.
- [108] H. Zhang, S. Zhang, H. Yuan, G. Lyu, J. Xie, FeCl₃-catalyzed ethanol pretreatment of sugarcane bagasse boosts sugar yields with low enzyme loadings and short hydrolysis time, *Bioresour. Technol.* 249 (2018) 395–401.
- [109] Y. Gao, J. Xu, Y. Zhang, Q. Yu, Z. Yuan, Y. Liu, Effects of different pretreatment methods on chemical composition of sugarcane bagasse and enzymatic hydrolysis, *Bioresour. Technol.* 144 (2013) 396–400.
- [110] S.H. Da Cruz, B.S. Dien, N.N. Nichols, B.C. Saha, M.A. Cotta, Hydrothermal pretreatment of sugarcane bagasse using response surface methodology improves digestibility and ethanol production by SSF, *J. Ind. Microbiol. Biotechnol.* 39 (2012) 439–447.
- [111] M. Zeng, E. Ximenes, M.R. Ladisch, N.S. Mosier, W. Vermerris, C.P. Huang, D.M. Sherman, Tissue-specific biomass recalcitrance in corn stover pretreated with

- liquid hot-water: enzymatic hydrolysis (part 1), *Biotechnol. Bioeng.* 109 (2012) 390–397.
- [112] J.C. García, M.A. Zamudio, A. Pérez, F. López, J.L. Colodette, Search for optimum conditions of Paulownia autohydrolysis process and influence in pulping process, *Environ. Prog. Sustain. Energy* 30 (2011) 92–101.
- [113] R.I. Ladeira-Azar, T. Morgan, G.P. Maitan-Alfenas, V.M. Guimarães, Inhibitors compounds on sugarcane bagasse saccharification: effects of pretreatment methods and alternatives to decrease inhibition, *Appl. Biochem. Biotechnol.* 188 (2019) 29–42.
- [114] Z. Hongdan, X. Shao-hua, W. Shubin, Enhancement of enzymatic saccharification of sugarcane bagasse by liquid hot water pretreatment, *Bioresour. Technol.* 143 (2013) 391–396.
- [115] E. Shirkavand, S. Baroutian, D.J. Gapes, B.R. Young, Combination of fungal and physicochemical processes for lignocellulosic biomass pretreatment—a review, *Renew. Sust. Energy Rev.* 54 (2016) 217–234.
- [116] C. Krishnan, L.D.C. Sousa, M. Jin, L. Chang, B.E. Dale, V. Balan, Alkali-based AFEX pretreatment for the conversion of sugarcane bagasse and cane leaf residues to ethanol, *Biotechnol. Bioeng.* 107 (2010) 441–450.
- [117] H. Zhang, S. Wu, Dilute ammonia pretreatment of sugarcane bagasse followed by enzymatic hydrolysis to sugars, *Cellulose* 21 (2014) 1341–1349.
- [118] A. Bala, B. Singh, Development of an environmental-benign process for efficient pretreatment and saccharification of *Saccharum* biomasses for bioethanol production, *Renew. Energy* 130 (2019) 12–24.
- [119] R. Liguori, V. Faraco, Biological processes for advancing lignocellulosic waste biorefinery by advocating circular economy, *Bioresour. Technol.* 215 (2016) 13–20.
- [120] D. Salvachúa, A. Prieto, M. López-Abelairas, T. Lu-Chau, A.T. Martínez, M.J. Martínez, Fungal pretreatment: an alternative in second-generation ethanol from wheat straw, *Bioresour. Technol.* 102 (2011) 7500–7506.
- [121] R. Masran, Z. Zanirun, E.K. Bahrin, M.F. Ibrahim, P.L. Yee, S. Abd-Aziz, Harnessing the potential of lignolytic enzymes for lignocellulosic biomass pretreatment, *Appl. Microbiol. Biotechnol.* 100 (2016) 5231–5246.
- [122] S. Chen, X. Zhang, D. Singh, H. Yu, X. Yang, Biological pretreatment of lignocelluloses: potential, progress and challenges, *Biofuels* 1 (2010) 177–199.
- [123] P. Kumar, D.M. Barrett, M.J. Delwiche, P. Stroeve, Methods for pretreatment of lignocellulosic biomass for efficient hydrolysis and biofuel production, *Ind. Eng. Chem. Res.* 48 (2009) 3713–3729.
- [124] L.D. Khuang, R. Kondo, R. De Leon, T.K. Anh, S. Meguro, K. Shimizu, I. Kamei, Effect of chemical factors on integrated fungal fermentation of sugarcane bagasse for ethanol production by a white-rot fungus, *Phlebia* sp. MG-60, *Bioresour. Technol.* 167 (2014) 33–40.
- [125] A. Jiraprasertwong, E. Gulari, S. Chavadej, Effect of different microbial strains on biological pretreatment of sugarcane bagasse for enzymatic hydrolysis, *World Acad. Sci. Eng. Technol. Int. J. Bioeng. Life Sci.* 8 (2014) 968–972.
- [126] A. da Silva Machado, A. Ferraz, Biological pretreatment of sugarcane bagasse with basidiomycetes producing varied patterns of biodegradation, *Bioresour. Technol.* 225 (2017) 17–22.
- [127] J. Rencoret, A. Pereira, J.C. del Río, A.T. Martínez, A. Gutiérrez, Delignification and saccharification enhancement of sugarcane by products by a laccase-based pretreatment, *ACS Sustain. Chem. Eng.* 5 (2017) 7145–7154.
- [128] D.D.S. Dos Santos, A.C. Camelo, K.C.P. Rodrigues, L.C. Carlos, N. Pereira, Ethanol production from sugarcane bagasse by *Zymomonas mobilis* using simultaneous saccharification and fermentation (SSF) process, *Appl. Biochem. Biotechnol.* 161 (2010) 93–105.
- [129] X. Zhao, Y. Song, D. Liu, Enzymatic hydrolysis and simultaneous saccharification and fermentation of alkali/peracetic acid-pretreated sugarcane bagasse for ethanol and 2,3-butanediol production, *Enzym. Microb. Technol.* 49 (2011) 413–419.
- [130] I. Amores, I. Ballesteros, P. Manzanares, F. Sáez, G. Michelena, M. Ballesteros, Ethanol production from sugarcane bagasse pretreated by steam explosion, *Elect. J. Energy Environ.* 1 (2013) 25–36.
- [131] S.K. Wahono, V.T. Rosyida, C. Darsih, D. Pratiwi, A. Frediansyah, Optimization of simultaneous saccharification and fermentation incubation time using cellulase enzyme for sugarcane bagasse on the second-generation bioethanol production technology, *Energy Procedia* 65 (2015) 333–336.
- [132] N. Sritrakul, S. Nitisinprasert, S. Keawsompong, Evaluation of dilute acid pretreatment for bioethanol fermentation from sugarcane bagasse pith, *Agric. Nat. Resour.* 51 (2017) 512–519.
- [133] Y. Jugwanth, Y. Sewsynker-Sukai, E.G. Kana, Valorization of sugarcane bagasse for bioethanol production through simultaneous saccharification and fermentation: optimization and kinetic studies, *Fuel* 262 (2020), 116552, .
- [134] A. de Araujo Guilherme, P.V.F. Dantas, C.E. de Araújo Padilha, E.S. dos Santos, G.R. de Macedo, Ethanol production from sugarcane bagasse: use of different fermentation strategies to enhance an environmental-friendly process, *J. Environ. Manag.* 234 (2019) 44–51.
- [135] K. Raj, C. Krishnan, Improved co-production of ethanol and xylitol from low-temperature aqueous ammonia pretreated sugarcane bagasse using two-stage high solids enzymatic hydrolysis and *Candida tropicalis*, *Renew. Energy* 153 (2020) 392–403.
- [136] P. Rattanapoltee, P. Kaewkannetra, Utilization of agricultural residues of pineapple peels and sugarcane bagasse as cost-saving raw materials in *Scenedesmus acutus* for lipid accumulation and biodiesel production, *Appl. Biochem. Biotechnol.* 173 (2014) 1495–1510.
- [137] Y. Song, Q. Li, X. Zhao, Y. Sun, D. Liu, Production of 2, 3-butanediol by *Klebsiella pneumoniae* from enzymatic hydrolyzate of sugarcane bagasse, *BioResources* 7 (2012) 4517–4530.
- [138] H. Su, G. Liu, M. He, F. Tan, A biorefining process: sequential, combinational lignocellulose pretreatment procedure for improving biobutanol production from sugarcane bagasse, *Bioresour. Technol.* 187 (2015) 149–160.
- [139] H. Li, L. Xiong, X. Chen, C. Wang, G. Qi, C. Huang, M. Luo, X. Chen, Enhanced enzymatic hydrolysis and acetone-butanol-ethanol fermentation of sugarcane bagasse by combined diluted acid with oxidate ammonolysis pretreatment, *Bioresour. Technol.* 228 (2016) 257–263.
- [140] P. Plangklang, A. Reungsang, S. Pattra, Enhanced bio-hydrogen production from sugarcane juice by immobilized *Clostridium butyricum* on sugarcane bagasse, *Int. J. Hydrog. Energy* 37 (2012) 15525–15532.
- [141] Z. Lai, M. Zhu, X. Yang, J. Wang, S. Li, Optimization of key factors affecting hydrogen production from sugarcane bagasse by a thermophilic anaerobic pure culture, *Biotechnol. Biofuels* 7 (2014) 119.
- [142] R.P. Ratti, T.P. Delforno, I.K. Sakamoto, M.B.A. Varesche, Thermophilic hydrogen production from sugarcane bagasse pretreated by steam explosion and alkaline delignification, *Int. J. Hydrog. Energy* 40 (2015) 6296–6306.
- [143] D. Jiang, Z. Fang, S.X. Chin, X.F. Tian, T.C. Su, Biohydrogen production from hydrolysates of selected tropical biomass wastes with *Clostridium butyricum*, *Sci. Rep.* 6 (2016) 1–11.
- [144] G. Prakash, A.J. Varma, A. Prabhune, Y. Shouche, M. Rao, Microbial production of xylitol from D-xylose and sugarcane bagasse hemicellulose using newly isolated thermotolerant yeast *Debaryomyces hansenii*, *Bioresour. Technol.* 102 (2011) 3304–3308.
- [145] S. Kamat, M. Khot, S. Zinjarde, A. RaviKumar, W.N. Gade, Coupled production of single cell oil as biodiesel feedstock, xylitol and xylanase from sugarcane bagasse in a biorefinery concept using fungi from the tropical mangrove wetlands, *Bioresour. Technol.* 135 (2013) 246–253.
- [146] J.F. Castañón-Rodríguez, J.M. Domínguez-González, B. Ortíz-Muñiz, B. Torrestiana-Sánchez, J.A.R. de León, M.G. Aguilar-Uscanga, Continuous multistep versus fed-batch production of ethanol and xylitol in a simulated medium of sugarcane bagasse hydrolysates, *Eng. Life Sci.* 15 (2015) 96–107.
- [147] L. Xu, L. Liu, S. Li, W. Zheng, Y. Cui, R. Liu, W. Sun, Xylitol production by *Candida tropicalis* 31949 from sugarcane bagasse hydrolysate, *Sugar Tech* 21 (2019) 341–347.
- [148] W.G. Moraes Junior, T.F. Pacheco, D. Trichez, J.R. Almeida, S.B. Gonçalves, Xylitol production on sugarcane biomass hydrolysate by newly identified *Candida tropicalis* JA2 strain, *Yeast* 36 (2019) 349–361.
- [149] D. Kumar, V.K. Jain, G. Shanker, A. Srivastava, Citric acid production by solid state fermentation using sugarcane bagasse, *Process Biochem.* 38 (2003) 1731–1738.
- [150] N.A. Amenaghawon, S.O. Areguamen, N.T. Agbroko, S.E. Ogbiede, C.O. Okieimen, Modelling and statistical optimisation of citric acid production from solid state fermentation of sugar cane bagasse using *Aspergillus niger*, *Int. J. Sci.* 2 (2013) 56–62.
- [151] E.R. Borges, N. Pereira, Succinic acid production from sugarcane bagasse hemicellulose hydrolysate by *Actinobacillus succinogenes*, *J. Ind. Microbiol. Biotechnol.* 38 (2011) 1001–1011.
- [152] R. Paranthaman, S. Kumaravel, K. Singaravadevel, Bioprocessing of sugarcane factory waste to production of Itaconic acid, *Afr. Microbiol. Res.* 8 (2014) 1672–1675.
- [153] L. Peng, N. Xie, L. Guo, L. Wang, B. Yu, Y. Ma, Efficient open fermentative production of polymer-grade L-lactate from sugarcane bagasse hydrolysate by thermotolerant *Bacillus* sp. strain P38, *PLoS One* 9 (2014).
- [154] D. Wei, X. Liu, S.T. Yang, Butyric acid production from sugarcane bagasse hydrolysate by *Clostridium tyrobutyricum* immobilized in a fibrous-bed bioreactor, *Bioresour. Technol.* 129 (2013) 553–560.
- [155] F. Chen, X. Peng, H. Xu, D. Zhang, P. Ouyang, Propionic acid production in a plant fibrous-bed bioreactor with immobilized *Propionibacterium freudenreichii* CCTCC M207015, *J. Biotechnol.* 164 (2013) 202–210.
- [156] L. Zhu, P. Wei, J. Cai, X. Zhu, Z. Wang, L. Huang, Z. Xu, Improving the productivity of propionic acid with FBB-immobilized cells of an adapted acid-tolerant *Propionibacterium acidipropionici*, *Bioresour. Technol.* 112 (2012) 248–253.
- [157] M.N.C. da Cunha, J.C. dos Santos Nascimento, M.C. Souza-Motta, A.V.P. Albertini, C.A. Lima, D.D.A.V. Marques, A.L.F. Porto, Production of enzymes by filamentous fungus using sugarcane and sugarcane bagasse as substrate, *Rev. Bras. Biocienc.* (2013) 11.
- [158] I. Norazlina, N. Meenalosani, Production of xylanase by *Trichoderma* sp. via solid state culture using sugarcane bagasse, *Int. J. Energy Sci* 3 (2013) 99–105.
- [159] M. Irfan, M. Nadeem, Q. Syed, One-factor-at-a-time (OFAT) optimization of xylanase production from *Trichoderma viride*-IR05 in solid-state fermentation, *J. Radiat. Res. Appl. Sci.* 7 (2014) 317–326.
- [160] F.A. Alebiosu, A.K. Lawal, S.O. Olatope, Y.L. Suberu, K.A. Shittu, F.A. Orji, O.D. Adelaja, A.I. Ugbanu, O.A. Esan, E.N. Dike, G.N. Elemo, Production of xylanases from fungal isolates by solid state fermentation using sugar cane bagasse, *Int. J. Curr. Microbiol. App. Sci.* 4 (2015) 323–331.
- [161] M. Irfan, U. Asghar, M. Nadeem, R. Nelofer, Q. Syed, Optimization of process parameters for xylanase production by *Bacillus* sp. in submerged fermentation, *J. Radiat. Res. Appl. Sci.* 9 (2016) 139–147.
- [162] C.G. Moraes, L.M. Sena, M.R. Lopes, A.R.O. Santos, K.O. Barros, C.R. Alves, A.P.T. Uetanabaro, M.A. Lachance, C.A. Rosa, Production of ethanol and xylanolytic enzymes by yeasts inhabiting rotting wood isolated in sugarcane bagasse hydrolysate, *Fungal Biol.* 124 (7) (2020) 639–647.
- [163] Alokika, B. Singh, Enhanced production of bacterial xylanase and its utility in saccharification of sugarcane bagasse, *Bioprocess Biosyst. Eng.* (2020) 1–11.
- [164] B. Benoliel, F.A.G. Torres, L.M.P. de Moraes, A novel promising *Trichoderma harzianum* strain for the production of a cellulolytic complex using sugarcane bagasse in natura, *Springer Plus* 2 (2013) 656.

- [165] B.M.P. Pereira, T.M. Alvarez, P. da Silva Delabona, A.J.P. Dillon, F.M. Squina, J.G. da Cruz Pradella, Cellulase on-site production from sugar cane bagasse using *Penicillium echinulatum*, *Bioenergy Res* 6 (2013) 1052–1062.
- [166] S.B. Onofre, G.C. Silva, S.P. Mattiello, D. Groth, I. Malagi, Production of cellulolytic enzymes by *Aspergillus flavus* using solid state fermentation based on sugarcane bagasse, *Am. J. Biochem.* 3 (2013) 25–28.
- [167] G.S.B. Salomão, J.C. Agnezi, L.B. Paulino, L.B. Hencker, T.S. de Lira, P.W. Tardioli, L. Pinotti, Production of cellulases by solid state fermentation using natural and pretreated sugarcane bagasse with different fungi, *Biocatal. Agric. Biotechnol* 17 (2018) 1–6.
- [168] G. Rajagopalan, C. Krishnan, α -Amylase production from catabolite derepressed *Bacillus subtilis* KCC103 utilizing sugarcane bagasse hydrolysate, *Bioresour. Technol.* 99 (2008) 3044–3050.
- [169] P. Rathakrishnan, P. Nagarajan, R. Rajeshkannan, Optimization of protease production by *Bacillus licheniformis* in sugarcane bagasse using statistical experimental design, *Res. Biotechnol.* (2012) 3.
- [170] Z. Vaseghi, G.D. Najafpour, S. Mohseni, S. Mahjoub, Production of active lipase by *Rhizopus oryzae* from sugarcane bagasse: solid state fermentation in a tray bioreactor, *Int. J. Food Sci. Technol.* 48 (2013) 283–289.
- [171] S.H.A. Naqvi, M.U. Dahot, M.Y. Khan, J.H. Xu, M. Rafiq, Usage of sugar cane bagasse as an energy source for the production of lipase by *Aspergillus fumigatus*, *Pak. J. Bot.* 45 (2013) 279–284.
- [172] K. Okano, S. Fukui, R. Kitao, T. Usagawa, Effects of culture length of *Pleurotuseryngii* grown on sugarcane bagasse on in vitro digestibility and chemical composition, *Anim. Feed Sci. Technol.* 136 (2007) 240–247.
- [173] S. Samadi, S. Wajizah, Y. Usman, D. Riyatsyah, Z. Al Firdausy, Improving sugarcane bagasse as animal feed by ammoniation and followed by fermentation with *Trichoderma harzianum* (in vitro study), *Anim. Prod.* 18 (2016) 14–21.
- [174] R. Lunsin, S. Duanyai, R. Pilajun, S. Duanyai, P. Sombatsri, Effect of urea- and molasses-treated sugarcane bagasse on nutrient composition and in vitro rumen fermentation in dairy cows, *Agric. Nat. Resour.* 52 (2018) 622–627.
- [175] V.C. Dos Santos, J.V. De Souza, C.R. Tarley, J. Caetano, D.C. Dragunski, Copper ions adsorption from aqueous medium using the biosorbent sugarcane bagasse in natura and chemically modified, *Water Air Soil Pollut.* 216 (2011) 351–359.
- [176] J. Bragatto, F. Segato, F.M. Squina, Production of xylooligosaccharides (XOS) from delignified sugarcane bagasse by peroxide-HAC process using recombinant xylanase from *Bacillus subtilis*, *Ind. Crop. Prod.* 51 (2013) 123–129.
- [177] F. Mandelli, L.B. Brenelli, R.F. Almeida, R. Goldbeck, L.D. Wolf, Z.B. Hoffmann, R. Ruller, G.J.M. Rocha, A.Z. Mercadante, F.M. Squina, Simultaneous production of xylooligosaccharides and antioxidant compounds from sugarcane bagasse via enzymatic hydrolysis, *Ind. Crop. Prod.* 52 (2014) 770–775.
- [178] X. Zhou, Y. Xu, Integrative process for sugarcane bagasse biorefinery to co-produce xylooligosaccharides and gluconic acid, *Bioresour. Technol.* 282 (2019) 81–87.
- [179] P.F. Ávila, J.P.L.F. Cairo, A. Damasio, M.B. Forte, R. Goldbeck, Xylooligosaccharides production from a sugarcane biomass mixture: effects of commercial enzyme combinations on bagasse/straw hydrolysis pretreated using different strategies, *Food Res. Int.* 128 (2020), 108702.
- [180] J.N. Nigam, Cultivation of *Candida langeronii* in sugar cane bagasse hemicellulosic hydrolyzate for the production of single cell protein, *World J. Microbiol. Biotechnol.* 16 (2000) 367–372.
- [181] S. Samadi, M. Mohammadi, G. Najafpour Darzi, Production of single cell protein from sugarcane bagasse by *Saccharomyces cerevisiae* in tray bioreactor, *Int. J. Eng. Sci.* 29 (2016) 1029–1036.
- [182] J.D.C.P. Scarpa, N.P. Marques, D.A. Monteiro, G.M. Martins, A.V. de Paula, M. Boscolo, R. da Silva, E. Gomes, D.A. Bocchini, Saccharification of pretreated sugarcane bagasse using enzymes solution from *Pycnoporus sanguineus* MCA 16 and cellulosic ethanol production, *Ind. Crop. Prod.* 141 (2019), 111795.
- [183] K. Kapoor, A.K. Tyagi, R.K. Diwan, Effect of gamma irradiation on recovery of total reducing sugars from delignified sugarcane bagasse, *Radiat. Phys. Chem.* 170 (2020), 108643.
- [184] M.K. Mishra, N. Khare, A.B. Agrawal, Bagasse cogeneration in India: status, barriers, *IOSR J. Mech. Civil Eng. (IOSR-JMCE)* 2 (2014) 69–78.
- [185] P.V. Neves, A.P. Pitarello, L.P. Ramos, Production of cellulosic ethanol from sugarcane bagasse by steam explosion: effect of extractives content, acid catalysis and different fermentation technologies, *Bioresour. Technol.* 208 (2016) 184–194.
- [186] P. Mishra, S. Krishnan, S. Rana, L. Singh, M. Sakinah, Z. Ab Wahid, Outlook of fermentative hydrogen production techniques: an overview of dark, photo and integrated dark-photo fermentative approach to biomass, *Energy Strategy Rev* 24 (2019) 27–37.
- [187] W. Cao, L. Guo, X. Yan, D. Zhang, X. Yao, Assessment of sugarcane bagasse gasification in supercritical water for hydrogen production, *Int. J. Hydrog. Energy* 43 (2018) 13711–13719.
- [188] M. Weissgram, C. Herwig, H.K. Weber, Biotechnological generation of value added products from spent pulping liquors: assessing the potential of extremophiles, *J. Bioprocess Biotech.* 5 (2015) 1–10.
- [189] S. Niju, M. Swathika, Delignification of sugarcane bagasse using pretreatment strategies for bioethanol production, *Biocatal. Agric. Biotechnol.* 101263 (2019).
- [190] M. Iram, U. Asghar, M. Irfan, Z. Huma, S. Jamil, M. Nadeem, Q. Syed, Production of bioethanol from sugarcane bagasse using yeast strains: a kinetic study, *Energy Sources, Part A: Recovery, Utilization, and Environmental Effects* 40 (2018) 364–372.
- [191] R.C. de Cerqueira Leite, M.R.L.V. Leal, L.A.B. Cortez, W.M. Griffin, M.I.G. Scandiffio, Can Brazil replace 5% of the 2025 gasoline world demand with ethanol? *Energy* 34 (2009) 655–661.
- [192] S.C. Nakanishi, L.B. Soares, L.E. Biazzi, V.M. Nascimento, A.C. Costa, G.J.M. Rocha, J.L. Ienczak, Fermentation strategy for second generation ethanol production from sugarcane bagasse hydrolyzate by *Spathaspora passalidarum* and *Scheffersomyces stipites*, *Biotechnol. Bioeng.* 114 (2017) 2211–2221.
- [193] N.R. Baral, L. Slutzky, A. Shah, T.C. Ezeji, K. Cornish, A. Christy, Acetone-butanol-ethanol fermentation of corn stover: current production methods, economic viability and commercial use, *FEMS Microbiol. Lett.* 363 (2016).
- [194] Z.W. Pang, W. Lu, H. Zhang, Z.W. Liang, J.J. Liang, L.W. Du, C.J. Duan, J.X. Feng, Butanol production employing fed-batch fermentation by *Clostridium acetobutylicum* GX01 using alkali-pretreated sugarcane bagasse hydrolysed by enzymes from *Thermoascus aurantiacus* QS 7-2-4, *Bioresour. Technol.* 212 (2016) 82–91.
- [195] R.S. Prakasham, R.S. Rao, P.J. Hobbs, Current trends in biotechnological production of xylitol and future prospects, *Curr. Trends Biotechnol. Pharm.* 3 (2009) 8–36.
- [196] D.T. Santos, B.F. Sarrouh, J.D. Rivaldi, A. Converti, S.S. Silva, Use of sugarcane bagasse as biomaterial for cell immobilization for xylitol production, *J. Food Eng.* 86 (2008) 542–548.
- [197] A.M. Contreras, E. Rosa, M. Pérez, H. Van Langenhove, J. Dewulf, Comparative life cycle assessment of four alternatives for using by-products of cane sugar production, *J. Clean. Prod.* 17 (2009) 772–779.
- [198] D.A.L. Silva, I. Delai, M.L.D. Montes, A.R. Ometto, Life cycle assessment of the sugarcane bagasse electricity generation in Brazil, *Renew. Sust. Energ. Rev.* 32 (2014) 532–547.
- [199] V.E.N. Santos, R.N. Ely, A.S. Szklo, A. Magrini, Chemicals, electricity and fuels from biorefineries processing Brazil's sugarcane bagasse: production recipes and minimum selling prices, *Renew. Sust. Energ. Rev.* 53 (2016) 1443–1458.
- [200] C.C. de Souza, J.P. Leandro, J.F. dos Reis Neto, D.M. Frainer, R.A. Castela, Cogeneration of electricity in sugar-alcohol plant: perspectives and viability, *Renew. Sust. Energ. Rev.* 91 (2018) 832–837.
- [201] D. Nyko, J.L.F. Garcia, A.Y. Milanez, N.J.D. Castro, R. Brandão, G.D.A. Dantas, Determinantes do baixo aproveitamento do potencial elétrico do setor sucroenergético: uma pesquisa de campo, *BNDES Setorial* 33 (2011) 421–476.
- [202] A. Gongora, D. Villafraanco, Sugarcane bagasse cogeneration in Belize: a review, *Renew. Sust. Energ. Rev.* 96 (2018) 58–63.
- [203] M. Wanapat, N. Anantasook, P. Rowlinson, R. Pilajun, P. Gunun, Effect of carbohydrate sources and levels of cotton seed meal in concentrate on feed intake, nutrient digestibility, rumen fermentation and microbial protein synthesis in young dairy bulls, *Asian-australas. J. Anim. Sci.* 26 (2013) 529–536.
- [204] N. Gunun, P. Gunun, M. Wanapat, A. Cherdthong, S. Kang, S. Polyorch, Improving the quality of sugarcane bagasse by urea and calcium hydroxide on gas production, degradability and rumen fermentation characteristics, *J. Anim. Plant Sci.* 27 (2017) 1758–1765.
- [205] N. Gunun, M. Wanapat, P. Gunun, A. Cherdthong, P. Khejornsart, S. Kang, Effect of treating sugarcane bagasse with urea and calcium hydroxide on feed intake, digestibility, and rumen fermentation in beef cattle, *Trop. Anim. Health Prod.* 48 (2016) 1123–1128.
- [206] F.N.A. Ferreira, W.M. Ferreira, C.S.S. Neta, D.F. da Silva Inácio, K.C. das Neves Mota, M.B. da Costa Júnior, L.F. da Rocha, L.B. Lara, D. de Oliveira Fontes, Effect of dietary inclusion of dried or autoclaved sugarcane bagasse and vinasse on live performance and in vitro evaluations on growing rabbits, *Anim. Feed Sci. Technol.* 230 (2017) 87–95.
- [207] M.H. Ahmed, S.A. Babiker, A.E.M.A. Fadel Elseed, A.M. Mohammed, Effect of urea-treatment on nutritive value of sugarcane bagasse, *ARPN J. Sci. Technol.* 3 (2013) 834–838.
- [208] A. Balgees, A. Elman, A.A. Hemeedan, R.I. Ahmed, Influence of different treatments on nutritive values of sugarcane bagasse, *Global Journal of Animal Scientific Research* 2 (2015) 295.
- [209] K. Sharma, S. Kuhar, R. Kuhad, P. Bhat, Combinatorial Approaches to Improve Plant Cell Wall Digestion: Possible Solution for Cattle Feed Problems, *IK International Publishing House Pvt. Ltd., New Delhi*, 2007.
- [210] X. Li, R. Kondo, K. Sakai, Biodegradation of sugarcane bagasse with marine fungus *Phlebia* sp. MG-60, *J. Wood Sci.* 48 (2002) 159–162.
- [211] M.M.M. Kermani, S. Bahrololoum, F. Koohzadi, Investigating the possibility of producing animal feed from sugarcane bagasse using oyster mushrooms: a case in rural entrepreneurship, *J. Global Entrepreneurship Res.* 9 (2019) 1–8.
- [212] Alokika, B. Singh, Production, characteristics, and biotechnological applications of microbial xylanases, *Appl. Microbiol. Biotechnol.* 103 (2019) 8763–8784.
- [213] N. Darabzadeh, Z. Hamidi-Esfahani, P. Hejazi, Optimization of cellulase production under solid-state fermentation by a new mutant strain of *Trichoderma reesei*, *FOOD Sci. Nutr.* 7 (2019) 572–578.
- [214] R. Sindhu, E. Gnansounou, P. Binod, A. Pandey, Bioconversion of sugarcane crop residue for value added products—an overview, *Renew. Energy* 98 (2016) 203–215.
- [215] I. Jain, V. Kumar, T. Satyanarayana, Xylooligosaccharides: an economical prebiotic from agroresidues and their health benefits, *Indian J. Exp. Biol.* 53 (2015) 131–142.
- [216] H. Li, X. Chen, L. Xiong, M. Luo, X. Chen, C. Wang, C. Huang, X. Chen, Stepwise enzymatic hydrolysis of alkaline oxidation treated sugarcane bagasse for the co-production of functional xylo-oligosaccharides and fermentable sugars, *Bioresour. Technol.* 275 (2019) 345–351.
- [217] C. Amorim, S.C. Silvério, K.L.J. Prather, L.R. Rodrigues, From lignocellulosic residues to market: production and commercial potential of xylooligosaccharides, *Biotechnol. Adv.* 37 (7) (2019) 107397.
- [218] L.A.R. Batalha, Q. Han, H. Jameel, H.M. Chang, J.L. Colodette, F.J.B. Gomes, Production of fermentable sugars from sugarcane bagasse by enzymatic hydrolysis after autohydrolysis and mechanical refining, *Bioresour. Technol.* 180 (2015) 97–105.
- [219] A. Anukam, S. Mamphweli, E. Meyer, O. Okoh, Computer simulation of the mass and energy balance during gasification of sugarcane bagasse, *J. Energy.* (2014) 1–9.

- [220] G. Akay, C.A. Jordan, Gasification of fuel cane bagasse in a downdraft gasifier: influence of lignocellulosic composition and fuel particle size on syngas composition and yield, *Energy Fuel* 25 (2011) 2274–2283.
- [221] A. Wazeer, Structural analysis of sugarcane bagasse as a feedstock in downdraft gasifier system—a review, *Int. J. Res. Eng. Innov.* 1 (2017) 223–228.
- [222] J.S. Piccin, C.S. Gomes, L.A. Feris, M. Gutterres, Kinetics and isotherms of leather dye adsorption by tannery solid waste, *Chem. Eng. J.* 183 (2012) 30–38.
- [223] C.A. Rezende, M.A. de Lima, P. Maziero, E.R. deAzevedo, W. Garcia, I. Polikarpov, Chemical and morphological characterization of sugarcane bagasse submitted to a delignification process for enhanced enzymatic digestibility, *Biotechnol. Biofuels* 4 (2011) 54.
- [224] A. Moubarik, N. Grimi, Valorization of olive stone and sugar cane bagasse by-products as biosorbents for the removal of cadmium from aqueous solution, *Food Res. Int.* 73 (2015) 169–175.
- [225] T.S. Fasoto, J.O. Arawande, A. Akinnusotu, Adsorption of zinc and chromium ions from aqueous solution onto sugarcane bagasse, *Int. J. Mod. Chem.* 6 (2014) 28–47.
- [226] P.L. Homagai, K.N. Ghimire, K. Inoue, Adsorption behavior of heavy metals onto chemically modified sugarcane bagasse, *Bioresour. Technol.* 101 (2010) 2067–2069.
- [227] F.O. Okafor, D.E. Ewa, Predicting the compressive strength of obudu earth blocks stabilized with cement kiln dust, *Niger. J. Technol.* 31 (2012) 149–155.
- [228] F. Adogla, P.P.K. Yalley, M. Arkoh, Improving compressed laterite bricks using powdered eggshells, *Int. J. Eng. Sci.* 5 (2016) 65–70.
- [229] A. Sales, S.A. Lima, Use of Brazilian sugarcane bagasse ash in concrete as sand replacement, *Waste Manag.* 30 (2010) 1114–1122.
- [230] K.C.P. Faria, R.F. Gurgel, J.N.F. Holanda, Recycling of sugarcane bagasse ash waste in the production of clay bricks, *J. Environ. Manag.* 101 (2012) 7–12.
- [231] S.R. Teixeira, A.E. Souza, C.L. Carvalho, V.C. Reynoso, M. Romero, J.M. Rincón, Characterization of a wollastonite glass-ceramic material prepared using sugar cane bagasse ash (SCBA) as one of the raw materials, *Mater. Charact.* 98 (2014) 209–214.
- [232] J. Torres Agredo, R. Mejía de Gutiérrez, C.E. Escandón Giraldo, L.O. González Salcedo, Characterization of sugar cane bagasse ash as supplementary material for Portland cement, *Ing. e Investig.* 34 (2014) 5–10.
- [233] M.A.S. Schettino, J.N.F. Holanda, Characterization of sugarcane bagasse ash waste for its use in ceramic floor tile, *Procedia Mater. Sci.* 8 (Supplement C) (2015) 190–196.
- [234] R.A. Naibaho, A. Rohana, S. Panggabean, Utilization of bagasse ash to reduce the use of cement in brick making, *Jurnal Rekayasa Pangan dan Pertanian* 3 (2015) 537–541.
- [235] R.W. Salim, J.M. Ndamuki, D.A. Adedokun, Improving the bearing strength of sandy loam soil compressed earth block bricks using sugarcane bagasse ash, *Sustainability* 6 (2014) 3686–3696.
- [236] A.P. Singh, P. Kumar, Light weight cement-sand and bagasse ash bricks, *Int. J. Innov. Res. Sci. Technol.* 1 (2015) 284–287.
- [237] A. Ritala, S.T. Häkkinen, M. Toivari, M.G. Wiebe, Single cell protein—state-of-the-art, industrial landscape and patents 2001–2016, *Front. Microbiol.* 8 (2017) 2009.
- [238] M.M. Mekonnen, A.Y. Hoekstra, Water footprint benchmarks for crop production: a first global assessment, *Ecol. Indic.* 46 (2014) 214–223.
- [239] M.D.O. Finco, L.D.G. Mamani, J.C.D. Carvalho, G.V. de Melo Pereira, V. Thomaz-Soccol, C.R. Soccol, Technological trends and market perspectives for production of microbial oils rich in omega-3, *Crit. Rev. Biotechnol.* 37 (2017) 656–671.
- [240] M. Yadegary, A. Hamidi, S.A. Alavi, E. Khodaverdi, H. Yahaghi, S. Sattari, G. Bagherpour, E. Yahaghi, Citric acid production from sugarcane bagasse through solid state fermentation method using *Aspergillus niger* mold and optimization of citric acid production by Taguchi method, *Jundishapur J. Microbiol.* 6 (2013).
- [241] D. Wischral, J.M. Arias, L.F. Modesto, D. de FrançaPassos, N. Pereira Jr., Lactic acid production from sugarcane bagasse hydrolysates by *Lactobacillus pentosus*: integrating xylose and glucose fermentation, *Biotechnol. Prog.* 35 (2019), e2718.
- [242] P. Chen, S. Tao, P. Zheng, Efficient and repeated production of succinic acid by turning sugarcane bagasse into sugar and support, *Bioresour. Technol.* 211 (2016) 406–413.
- [243] H. Zhang, J. Xu, X. Su, J. Bao, K. Wang, Z. Mao, Citric acid production by recycling its wastewater treated with anaerobic digestion and nanofiltration, *Process Biochem.* 58 (2017) 245–251.
- [244] D.M. Chetan, Comparison of citric acid production from *Aspergillus niger* in solid and suspension state fermentation, *Int. J. Green Pharm* (2018) 12.
- [245] A.R. Angumeenal, D. Venkappayya, An overview of citric acid production, *LWT-Food Sci. Technol.* 50 (2013) 367–370.
- [246] G.S. Dhillon, S.K. Brar, S. Kaur, M. Verma, Bioproduction and extraction optimization of citric acid from *Aspergillus niger* by rotating drum type solid-state bioreactor, *Ind. Crop. Prod.* 41 (2013) 78–84.
- [247] R.G. Bastos, D.V. Morais, M.P.C. Volpi, Influence of solid moisture and bed height on cultivation of *Aspergillus niger* from sugarcane bagasse with vinasse, *Braz. J. Chem. Eng.* 32 (2015) 377–384.
- [248] A.F. Oliveira, V. de Carvalho Matos, R.G. Bastos, Cultivation of *Aspergillus niger* on sugarcane bagasse with vinasse, *Bioscience J* (2012) 28.
- [249] B.S. Campanhol, G.C. Silveira, M.C. Castro, S.R. Ceccato-Antonini, R.G. Bastos, Effect of the nutrient solution in the microbial production of citric acid from sugarcane bagasse and vinasse, *Biocatal. Agric. Biotechnol.* 19 (2019), 101147.
- [250] M.A. Abdel-Rahman, Y. Tashiro, K. Sonomoto, Recent advances in lactic acid production by microbial fermentation processes, *Biotechnol. Adv.* 31 (2013) 877–902.
- [251] D. Awasthi, L. Wang, M.S. Rhee, Q. Wang, D. Chauliac, L.O. Ingram, K.T. Shanmugam, Metabolic engineering of *Bacillus subtilis* for production of D-lactic acid, *Biotechnol. Bioeng.* 115 (2018) 453–463.
- [252] M. Karamanlioglu, R. Preziosi, G.D. Robson, Abiotic and biotic environmental degradation of the bioplastic polymer poly (lactic acid): a review, *Polym. Degrad. Stab.* 137 (2017) 122–130.
- [253] V.C. Ghantani, S.T. Lomate, M.K. Dongare, S.B. Umbarkar, Catalytic dehydration of lactic acid to acrylic acid using calcium hydroxyapatite catalysts, *Green Chem.* 15 (2013) 1211–1217.
- [254] F.A.C. Martinez, E.M. Balciunas, J.M. Salgado, J.M.D. González, A. Converti, R.P. de Souza Oliveira, Lactic acid properties, applications and production: a review, *Trends Food Sci. Technol.* 30 (2013) 70–83.
- [255] P. Unrean, Optimized feeding schemes of simultaneous saccharification and fermentation process for high lactic acid titer from sugarcane bagasse, *Ind. Crop. Prod.* 111 (2018) 660–666.
- [256] C. Edwin, E.C. van der Pol, G. Eggink, R.A. Weusthuis, Production of L (+)-lactic acid from acid pretreated sugarcane bagasse using *Bacillus coagulans* DSM2314 in a simultaneous saccharification and fermentation strategy, *Biotechnol. Biofuels* 9 (2016) 248.
- [257] D. Yamasaki, Y. Minouchi, M. Ashiuchi, Extremolyte-like applicability of an archaeal exopolymer, poly-γ-l-glutamate, *Environ. Technol.* 31 (2010) 1129–1134.
- [258] Y.H. Hsueh, K.Y. Huang, S.C. Kunene, T.Y. Lee, Poly-γ-glutamic acid synthesis, gene regulation, phylogenetic relationships, and role in fermentation, *Int. J. Mol. Sci.* 18 (2017) 2644.
- [259] A.J. Anju, P. Binod, A. Pandey, Production and characterization of microbial poly-γ-glutamic acid from renewable resources, *Indian J. Exp. Biol.* 55 (2017) 405–410.
- [260] X. Feng, B. Tang, Y. Jiang, Z. Xu, P. Lei, J. Liang, H. Xu, Efficient production of poly-γ-glutamic acid from cane molasses by *Bacillus subtilis* NX-2 immobilized on chemically modified sugarcane bagasse, *J. Chem. Technol. Biotechnol.* 91 (2016) 2085–2093.
- [261] A.A. Koutinas, A. Chatzifragkou, N. Kopsahelis, S. Papanikolaou, I.K. Kookos, Design and techno-economic evaluation of microbial oil production as a renewable resource for biodiesel and oleochemical production, *Fuel* 116 (2014) 566–577.
- [262] I.R. Sitepu, M. Jin, J.E. Fernandez, L. da Costa Sousa, V. Balan, K.L. Boundy-Mills, Identification of oleaginous yeast strains able to accumulate high intracellular lipids when cultivated in alkaline pretreated corn stover, *Appl. Microbiol. Biotechnol.* 98 (2014) 7645–7657.
- [263] P.J. Slininger, B.S. Dien, C.P. Kurtzman, B.R. Moser, E.L. Bakota, S.R. Thompson, P.J. O'Bryan, M.A. Cotta, V. Balan, M. Jin, L.D.C. Sousa, Comparative lipid production by oleaginous yeasts in hydrolysates of lignocellulosic biomass and process strategy for high titers, *Biotechnol. Bioeng.* 113 (2016) 1676–1690.
- [264] N. Bonturi, A. Crucello, A.J.C. Viana, E.A. Miranda, Microbial oil production in sugarcane bagasse hemicellulosic hydrolysate without nutrient supplementation by a *Rhodospiridium toruloides* adapted strain, *Process Biochem.* 57 (2017) 16–25.
- [265] C. Huang, X.F. Chen, L. Xiong, L.L. Ma, Y. Chen, Single cell oil production from low-cost substrates: the possibility and potential of its industrialization, *Biotechnol. Adv.* 31 (2013) 129–139.
- [266] T.K. Hari, Z. Yaakob, N.N. Binita, Aviation biofuel from renewable resources: routes, opportunities and challenges, *Renew. Sust. Energ. Rev.* 42 (2015) 1234–1244.
- [267] S. Papanikolaou, M. Rontou, A. Belka, M. Athenaki, C. Gardeli, A. Mallouchos, O. Kalantzi, A.A. Koutinas, I.K. Kookos, A.P. Zeng, G. Aggelis, Conversion of biodiesel-derived glycerol into biotechnological products of industrial significance by yeast and fungal strains, *Eng. Life Sci.* 17 (2017) 262–281.
- [268] M. Athenaki, C. Gardeli, P. Diamantopoulou, S.S. Tchakouteu, D. Sarris, A. Philippoussis, S. Papanikolaou, Lipids from yeasts and fungi: physiology, production and analytical considerations, *J. Appl. Microbiol.* 124 (2018) 336–367.
- [269] C. Dias, C. Silva, C. Freitas, A. Reis, T.L. Da Silva, Effect of medium pH on *Rhodospiridium toruloides* NCYC 921 carotenoid and lipid production evaluated by flow cytometry, *Appl. Biochem. Biotechnol.* 179 (2016) 776–787.
- [270] C. Zhang, H. Shen, X. Zhang, X. Yu, H. Wang, S. Xiao, J. Wang, Z.K. Zhao, Combined mutagenesis of *Rhodospiridium toruloides* for improved production of carotenoids and lipids, *Biotechnol. Lett.* 38 (2016) 1733–1738.
- [271] A.M. Kot, S. Błażej, A. Kurcz, I. Gientka, M. Kieliszek, *Rhodotorulaglutinis*—potential source of lipids, carotenoids, and enzymes for use in industries, *Appl. Microbiol. Biotechnol.* 100 (2016) 6103–6117.
- [272] G. Cai, L. Moghaddam, I.M. O'Hara, Z. Zhang, Microbial oil production from acidified glycerol pretreated sugarcane bagasse by *Mortierella isabellina*, *RSC Adv.* 9 (2019) 2539–2550.
- [273] M. Hassanpour, G. Cai, L.K. Gebbie, R.E. Speight, V.S. Junior Te'o, I.M. O'Hara, Z. Zhang, Co-utilization of acidified glycerol pretreated-sugarcane bagasse for microbial oil production by a novel *Rhodospiridium* strain, *Eng. Life Sci.* 19 (2019) 217–228.
- [274] S. Yadav, G. Gupta, R. Bhatnagar, A review on composition and properties of bagasse fibers, *Cellulose* 45 (2015) 55.
- [275] B. Parameswaran, Sugarcane bagasse, *Biotechnology for Agro-industrial Residues Utilisation*, Springer, Dordrecht 2009, pp. 239–252.
- [276] F.M. Giro, C. Fonseca, F. Carvalheiro, L.C. Duarte, S. Marques, R. Bogel-Lukasik, Hemicelluloses for fuel ethanol: a review, *Bioresour. Technol.* 101 (2010) 4775–4800.
- [277] A.M. Zetty-Arenas, R.F. Alves, C.A.F. Portela, A.P. Mariano, T.O. Basso, L.P. Tovar, R. Maciel Filho, S. Freitas, Towards enhanced n-butanol production from sugarcane bagasse hemicellulosic hydrolysate: strain screening, and the effects of sugar concentration and butanol tolerance, *Biomass Bioenergy* 126 (2019) 190–198.